

Source files received / data types found

Survivors

1. [Survivors_SP_2025-01-26](#)

SP / TMNT survivors: Raphael, Leonardo, Donatello, Michelangelo, April, Splinter. Includes yellow-star upgrades, red awakening upgrades, passive skills, active skills, teamwork passive slots, crit rate, laceration, vulnerability, pet boosts, etc.

Tech Part Resonance

2. [Tech_Part_Resonance_Drone](#)

Drone resonance: Augment Wingbit + Attack Wingbit, damage scaling, cooldown, duration, what gets boosted.

3. [Tech_Part_Resonance_HI_Maintainer_RPG](#)

C4 / RPG / HI-Maintainer style resonance. Shows base damage, launch interval, explosion delay, max values.

Equipment

4. [Weapons_V2_2](#)

Weapons: Void Power, Lightchaser, Sword of Disorder, Kunai, Baseball Bat, Katana, Shotgun, Revolver.

5. [Armor_V2](#)

Chest armor: Eternal Suit, Voidwaker Windbreaker, Armor of Quietus, Full Metal Suit, Traveler's Jacket, Carapace, Army Uniform, Protective Suit.

6. [Belts_V2](#)

Belts: Twisting Belt, Eternal Belt, Stylish Belt, Leather Belt, Voidwaker Sash, Army Belt, Plated Sensor, Broad Waistguard.

7. [Boots_V2](#)

Boots: Voidwaker Treads, Eternal Boots, Light Runners, Shoes of Confusion, Army Boots, Prosthetic Legs, Layered Snowshoes, Highboots.

Forge / Cores

8. [Astral_Forge](#)

AF requirements, SS gear requirements, non-SS AF costs, salvage returns, SS breakpoints.

9. [Awakening_Core_Guide](#)

Awakening core priority paths for Yang, Metallia, King, Worm, Common, Catnips, TMNT survivors in Ender's Echo and Clan Expedition.

Debuffs / Damage Multipliers

10. [Boss_Debuffs](#)

Chill, Poison, Exposed, Weaken, Debilitate, Yin Yang, Laceration, Vulnerability. This is very important for DPS stacking logic.

11. [Crit_Rate_And_Damage](#)

Crit rate and crit damage sources from survivor levels, gear, collectibles, skills, pets, talents, combat harmony, survivor synergy, Don passive, Rex, etc.

Progression / Unlocks

12. [Chapter_Unlocks](#)

Unlock timing by chapter: shop, equipment, tech parts, pets, clans, Ender's Echo, Zone Ops, Special Ops, privilege card, etc.

Clan

13. [Clan_Expedition_Rewards](#)

Novice, Adept, Elite clan expedition rewards: chapter rewards, chest rewards, ranking rewards, clan tier limits.

14. [Clan_Level_Rewards](#)

Clan level perks, clan EXP, clan size, quick earnings, clan member ATK/HP boosts.

15. [Clan_Shop](#)

Clan shop items, costs, quantities, and how stock changes by clan level.

Collectibles

16. [Collectible_Chest_Odds](#)

Collectible chest rarity odds and shard odds.

17. [Collectible_Costs](#)

Collectible star costs, shard costs, exchange rates, collector's hearts, purchase costs.

18. [Collectibles_Edition_1](#)

1st Edition collectible chest item pool and drop rates.

19. [Collectibles_Edition_2](#)

2nd Edition collectible chest item pool and drop rates.

20. [Collectibles_Edition_3](#)

3rd Edition collectible chest item pool and drop rates.

Data type 1: SP Survivors / TMNT Survivors

Source: [Survivors_SP_2025-01-26](#)

This image is about **SP-grade survivor progression**, specifically the TMNT collab survivors:

Raphael, Leonardo, Donatello, Michelangelo, April, Splinter

This data type belongs in the optimizer as:

Survivor | SP Survivor | Star Upgrade | Awakening | Passive Skill |
Active Skill | Teamwork Passive | Debuff Source | Crit Source | Pet
Source

For your calculator, this should mostly affect **DPS**, not HP. The HP lines should still be stored as raw data, but scoring should mark them as **ignore / non-DPS**.

Shared SP survivor structure

Each SP survivor has:

Yellow stars

- 1-star: unlocks exclusive weapon / base survivor skill.
- 2-star: Survivor License, usually **Survivor ATK +400** and **Survivor HP +2000**.
- 3-star: improves the survivor's main weapon or attack mechanic.
- 4-star: Advanced Survivor License, usually **Survivor ATK +1000** and **Survivor HP +5000**.
- 5-star: **All Survivors: Crit Rate +8%**.
- 6-star: big personal DPS upgrade, usually main weapon damage + survivor ATK +20%.

Red awakening stars

- Adds active awakening skill improvements.
- Adds passive awakening skills.
- Some give **Teamwork Passive Slots**, which matter a lot because awakened passive skills can be shared/equipped.
- Costs appear to scale with awakening cores, pizza/slices, and blue materials.

For calculator logic, **5 yellow stars is extremely important** because it gives global crit rate. The HP parts are mostly trash for your DPS optimizer.

Raphael

Main association

Raphael is associated with:

Sai / Fury Sai / Bleed / Laceration / Lacerated-target damage / Rocket Assault

Yellow-star upgrades

1-star

- Exclusive weapon: **Sai**
- Raphael throws **2 sai** at enemies.
- Can evolve into **Fury Sai**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Power Strike**
- For every 5 attacks, 1 more giant sai is thrown.
- Attacks inflict **Bleed**.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Hot-headed**
- Sai and Fury Sai damage +20%.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Rocket Assault**
- When Rocket Assault ends, fires a wave of **Assault Sai**.
- Passive: **Scattered Spikes**
- Throws piercing sai around every 20 seconds.
- Enemies hit get **+5 seconds of Laceration**.
- Damage to Lacerated enemies +10%.

A2

- Additional damage on Lacerated targets +10%.

A3

- More Assault Sai thrown during Rocket Assault.

A4

- Additional damage on Lacerated targets +10%.
- Unlocks 1 Teamwork Passive Slot.

A5

- Fires Assault Sai during Rocket Assault.

A6

- Additional damage on Lacerated targets +20%.
- Teamwork Passive Slot +1.

Optimizer notes

Raphael is a **Laceration damage amplifier**. He matters if the build uses Laceration stacking, boss debuffs, or teamwork passive sharing.

Database flags:

Debuff = Bleed, Laceration
DamageType = Lacerated target damage
GlobalValue = Crit Rate at 5-star
TeamworkSlot = A4 and A6
DPSRelevant = Yes

Leonardo

Main association

Leonardo is associated with:

Dual Katanas / Fullmoon Slash / Laceration / Whirlwind Slash /
Laceration duration

Yellow-star upgrades

1-star

- Exclusive weapon: **Swiftwind Dual Katanas**
- Leonardo performs long and short slashes with his katanas.
- Can evolve into **Fullmoon Slash**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Voidbreaker Combo**
- Adds long and short slashes.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Secret Art Limit**
- Fullmoon Slash breakthrough art.
- Adds attack.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Whirlwind Slash**
- Whirlwind Slash damage +50%.
- Passive: **Undaunted Will**
- When hit by monster collisions, inflicts Laceration.
- Damage on Lacerated targets +2% for 5 seconds.

A2

- Additional damage on Lacerated targets +2%.
- Laceration duration +5 seconds.

A3

- Reduces Whirlwind Slash cooldown.

A4

- Additional damage on Lacerated targets +3%.
- Laceration duration +5 seconds.
- Unlocks 1 Teamwork Passive Slot.

A5

- Whirlwind Slash duration up.

A6

- Additional damage on Lacerated targets +3%.
- Laceration duration becomes permanent.
- Teamwork Passive Slot +1.

Optimizer notes

Leonardo is another **Laceration support/DPS source**, but his numbers look smaller than Raphael/Michelangelo. His value is more about Laceration uptime and duration.

Database flags:

Debuff = Laceration

DamageType = Lacerated target damage

Mechanic = Duration extension / permanent laceration

GlobalValue = Crit Rate at 5-star

TeamworkSlot = A4 and A6

DPSRelevant = Yes

Donatello

Main association

Donatello is associated with:

Bo / Savage Bo / Skill Damage / Crit Damage / Crit Rate / SP passive scaling / Brotherhood

This is one of the most important ones for the calculator because Donatello has **global SP passive scaling**.

Yellow-star upgrades

1-star

- Exclusive weapon: **Seeker Bo**
- Donatello spins his bo to attack enemies.
- Can be upgraded to **Savage Bo**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Super Sweep**
- Seeker Bo and Savage Bo damage +30%.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- Staff Strike / Seeker Bo / Savage Bo:
 - Attack range up
 - Damage +30%
- Survivor ATK +20%

Awakening upgrades

A1

- Active: **Spinning Smash**
- Fires an energy wave when Spinning Smash ends.
- Passive: **Brotherhood**
- Exclusive global passive.
- Effective for all Survivors when unlocked.
- Cannot be equipped to Teamwork passive slots.
- For every SP Grade Survivor Awakening passive skill equipped:
 - Crit Rate +3%

A2

- Brotherhood adds:
 - Skill Damage +3%
 - Crit Damage +3%
 - Per SP-grade survivor awakening passive equipped.

A3

- Spinning Smash attack range up.

A4

- Brotherhood adds damage to special debuffed targets:
 - Damage to Lacerated targets +3%
 - Damage to Poisoned targets +3%

- Damage to Chilled targets +3%
- Unlocks 1 Teamwork Passive Slot.

A5

- Spinning Smash damage +50%.

A6

- Brotherhood adds:
 - Damage to Weakened targets +3%
 - Damage to Poisoned targets +3%
 - Damage to Chilled targets +3%
- Teamwork Passive Slot +1.

Optimizer notes

Donatello is a **global scaling engine**. He should not be treated like a normal survivor only. His Brotherhood passive creates value based on how many SP awakening passives are equipped.

Database flags:

GlobalPassive = Brotherhood
 ScalingInput = Equipped SP Awakening Passive Count
 CritRateSource = +3% per SP passive at A1
 SkillDamageSource = +3% per SP passive at A2
 CritDamageSource = +3% per SP passive at A2
 DebuffDamage = Lacerated / Poisoned / Chilled / Weakened
 TeamworkSlot = A4 and A6
 DPSRelevant = Very High

This one is important for your optimizer because it affects **build synergy**, not just one survivor.

Michelangelo

Main association

Michelangelo is associated with:

Nunchucks / Windchaser Nunchucks / Whirlwind / Bleed / Laceration / Lacerated-target damage / hit-count trigger

Yellow-star upgrades

1-star

- Exclusive weapon: **Arc Nunchucks**
- Swings nunchucks and launches whirlwinds.
- Can be upgraded to **Windchaser Nunchucks**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Tubular Tornado**
- Arc Nunchucks / Windchaser Nunchucks whirlwind damage +50%.
- Whirlwind range up.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Apocalypse Tornado**
- Arc Nunchucks / Windchaser Nunchucks whirlwind damage +50%.
- Whirlwind inflicts **Bleed**.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Twin Throw**
- Nunchucks thrown by Twin Throw can be thrown again.
- Passive: **Fantastic Storm**
- Every 500 attacks on hit spawns 1 windstorm.
- Windstorm inflicts **Lacerate** for 5 seconds.
- Skill cooldown: 20 seconds.
- Damage to Lacerated targets +10%.

A2

- Fantastic Storm:
 - Required hits -100
 - Damage to Lacerated targets +10%

A3

- Twin Throw thrown nunchucks count up.

A4

- Fantastic Storm:
 - Required hits -100
 - Damage to Lacerated targets +10%
- Unlocks 1 Teamwork Passive Slot.

A5

- Twin Throw nunchucks attack range up.
- Attack damage +50%.

A6

- Fantastic Storm:
 - Required hits -200
 - Damage to Lacerated targets +20%
- Teamwork Passive Slot +1.

Optimizer notes

Michelangelo looks like a stronger Laceration DPS support than Leonardo because he has bigger Lacerated-target damage chunks and hit-count scaling.

Database flags:

Debuff = Bleed, Laceration
Mechanic = Hit count trigger
Trigger = Every 500 attacks, reduced by awakenings
DamageType = Lacerated target damage
GlobalValue = Crit Rate at 5-star
TeamworkSlot = A4 and A6
DPSRelevant = High

April

Main association

April is associated with:

Soundwave Balls / Mega Soundwave Balls / Vulnerability / Pet ATK / Pet HP / Reporter Snapshot

April matters because she connects **survivor awakening + pet damage + vulnerability**.

Yellow-star upgrades

1-star

- Exclusive weapon: **Microphone**
- April fires soundwave balls at enemies.
- Can evolve into **Mega Soundwave Balls**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Increased Volume**
- Soundwave Ball damage +30%.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Sonic Boom**
- Mega Soundwave Balls explode in a small area when enemies are hit.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Reporter Snapshot**
- Reporter Snapshot damage +50%.
- Passive: **Pet Companion**
- When in battle, deployed pet's:
 - Attack +5%
 - HP +5%

A2

- Deployed pet additional:
 - Attack +5%
 - HP +5%

A3

- Reporter Snapshot damage +50%.
- Targets hit by Snapshot are inflicted with **5% Vulnerability**.

A4

- Deployed pet additional:
 - Attack +10%
 - HP +10%
- Unlocks 1 Teamwork Passive Slot.

A5

- Reporter Snapshot damage +50%.
- Targets hit by Snapshot get additional **5% Vulnerability**.

A6

- Deployed pet additional:
 - Attack +10%
 - HP +10%
- Teamwork Passive Slot +1.

Optimizer notes

April is not just personal damage. She is a **pet amplifier** and **vulnerability source**.

Database flags:

```

Debuff = Vulnerability
PetATKSource = Yes
PetHPSource = Ignore for DPS
Vulnerability = 5% at A3, additional 5% at A5
GlobalValue = Crit Rate at 5-star
TeamworkSlot = A4 and A6
DPSRelevant = High if pet build / vulnerability stacking
  
```

For your optimizer, April should become more valuable if the player uses a strong pet setup.

Splinter

Main association

Splinter is associated with:

```

Secret Art Blast / Secret Art Fusion / Slow / Enhanced Slow / Phantom
Assault / Pet Crit Rate / Pet Crit Damage / Pet Skill CD
  
```

Splinter is a pet-crit support survivor.

Yellow-star upgrades

1-star

- Exclusive weapon: **Secret Art Blast**
- Splinter brandishes his cane and fires a power wave at enemies.
- Can evolve into **Secret Art Fusion**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Bonecrusher Blast**
- Secret Art Blast / Secret Art Fusion power wave damage +30%.
- Inflicts **Slow**.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Fatality Blast**
- Secret Art Blast / Secret Art Fusion power wave damage +30%.
- Inflicts **enhanced Slow**.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Phantom Assault**
- Adds 1 phantom.
- Phantom damage +20%.
- Passive: **Master Mentor**
- Equipped pet skill cooldown -5%.
- Pet Crit Rate +10%.

A2

- Master Mentor:
 - Equipped Pet Crit Rate +10%
 - Pet Crit Damage +20%

A3

- Phantom Assault:
 - +1 Phantom
 - Phantom damage +20%

A4

- Master Mentor:
 - Equipped Pet Skill cooldown -15%
- Unlocks 1 Teamwork Passive Slot.

A5

- Phantom Assault:
 - +1 Phantom
 - Phantom damage +20%

A6

- Master Mentor:
 - Equipped Pet Skill cooldown -10%
 - Pet Crit Rate +20%
 - Pet Crit Damage +20%
- Teamwork Passive Slot +1.

Optimizer notes

Splinter is heavily tied to **pet crit scaling** and pet skill cooldown. He should become stronger in the calculator if pets are a major DPS source.

Database flags:

PetCritRateSource = Yes

PetCritDamageSource = Yes

PetSkillCooldownSource = Yes

Debuff = Slow / Enhanced Slow

GlobalValue = Crit Rate at 5-star

TeamworkSlot = A4 and A6
DPSRelevant = High for pet builds

Important optimizer associations from this whole source

1. Global crit from SP survivors

Every SP survivor shown has:

5 yellow stars = All Survivors Crit Rate +8%

That means each 5-star SP survivor is a **global DPS upgrade**, even if you do not use that survivor.

For the optimizer:

CritRateGlobal += 8% per SP survivor at 5 yellow stars

This is very important for your damage-only calculator.

2. Laceration package

Laceration is tied to:

- Raphael
- Leonardo
- Michelangelo
- Donatello Brotherhood scaling
- Boss Debuffs source image also has a Laceration section

This means Laceration should be treated as a real debuff family.

Suggested field:

DebuffFamily = Laceration
DamageModifierType = Damage to Lacerated targets
StackingType = Needs testing / source-specific

3. Vulnerability package

April adds Vulnerability through Reporter Snapshot.

This connects directly to the **Boss_Debuffs** source because Vulnerability is a major damage amplifier.

Suggested field:

DebuffFamily = Vulnerability
Source = April Awakening A3/A5
DPSRelevant = Very High

4. Pet package

April and Splinter both connect to pets.

April:

- Pet ATK
- Pet HP

Splinter:

- Pet Crit Rate
- Pet Crit Damage
- Pet Skill cooldown

For your calculator:

PetATK = count for DPS
PetCritRate = count for DPS
PetCritDamage = count for DPS
PetSkillCDReduction = count for DPS if pet skill damage is modeled
PetHP = store but ignore

5. Teamwork Passive Slots

Most SP survivors unlock Teamwork Passive Slots at awakening levels:

- A4: Unlocks 1 Teamwork Passive Slot
- A6: Teamwork Passive Slot +1

This matters because survivor awakening is not just “upgrade survivor.” It changes how many passive skills can be equipped and shared.

Suggested fields:

```
TeamworkPassiveSlotUnlock_A4 = 1  
TeamworkPassiveSlotUnlock_A6 = +1
```

Clean database format for this data type

Use something like this:

```
SurvivorID  
SurvivorName  
Grade  
CollabGroup  
StarType  
StarLevel  
UpgradeName  
UpgradeDescription  
AffectedStat  
StatAmount  
TargetScope  
AssociatedDebuff  
AssociatedSkill  
AssociatedPetStat  
TeamworkPassiveSlotChange  
DPSRelevant  
IgnoreForDPS  
Notes  
SourceImage  
SourceDate
```

Example row:

SP_RAPHAEL | Raphael | SP | TMNT | Yellow | 5 | All Survivors Crit Rate | Crit Rate +8% | CritRate | 8% | All Survivors | None | None | None | 0 | Yes | No | Global DPS upgrade | Survivors_SP_2025-01-26

Example row:

SP_APRIL | April | SP | TMNT | Awakening | A3 | Reporter Snapshot | Snapshot damage +50%; targets get 5% Vulnerability | Vulnerability | 5% | Enemy Hit By Snapshot | Vulnerability | Reporter Snapshot | None | 0 | Yes | No | Boss damage amplifier | Survivors_SP_2025-01-26

Next data type should be **Crit Rate/Crit Damage** or **Boss Debuffs**, because those explain how these survivor bonuses actually stack into DPS.

Data type 2: Crit Rate & Crit Damage

Source: [Crit_Rate_And_Damage_2025-03-09](#)

This image is a **stat-source registry**. It is not just showing one item. It shows every major place Crit Rate and Crit Damage can come from.

For the optimizer, this should become:

StatSource | CritRate | CritDamage | Global | Conditional | GearBased
| SurvivorBased | PetBased | SkillBased | CollectibleBased |
DPSRelevant

This is very important because crit is one of the biggest DPS multipliers in Survivor.io.

Main optimizer rule

Do **not** store crit as one single number only.

You need to split it into:

Global Crit Rate

Global Crit Damage

Active Survivor Crit Sources

Equipped Gear Crit Sources

In-battle Skill Crit Sources
Pet Crit Sources
Collectible Crit Sources
Conditional Crit Sources
Temporary / stance-based Crit Sources

Because some are always active, some require a specific survivor, some require a pet to be active, some require a gear condition, and some require in-battle setup.

Crit Rate section

1. Survivor level Crit Rate

This is associated with:

[Survivor Level Milestone](#) | [Global Stat](#) | [Account DPS](#)

The source shows **98% total global Crit Rate** possible from survivor levels / special survivor unlocks.

Normal survivor level sources shown

- King level 80: Crit Rate +5%
- Tsukuyomi level 80: Crit Rate +5%
- Worm level 80: Crit Rate +5%
- Wesson / other survivor icon level 120: Crit Rate +5%
- Master Yang level 120: Crit Rate +10%

SpongeBob event survivors

These are marked as special-event only.

- SpongeBob level 120: Crit Rate +5%
- Squidward level 120: Crit Rate +5%
- Patrick level 120: Crit Rate +5%
- Sandy level 120: Crit Rate +5%

TMNT SP survivors

Each TMNT survivor gives:

5 yellow stars = All Survivors Crit Rate +8%

Shown TMNT survivors:

- Raphael 5-star: Crit Rate +8%
- Leonardo 5-star: Crit Rate +8%
- April 5-star: Crit Rate +8%
- Michelangelo 5-star: Crit Rate +8%
- Donatello 5-star: Crit Rate +8%
- Splinter 5-star: Crit Rate +8%

Optimizer notes

This is a **global permanent DPS source**. It should have high value because it improves all builds, not just one survivor.

Database flag:

SourceCategory = SurvivorLevel

TargetScope = AllSurvivors

StatType = CritRate

DPSRelevant = Very High

ConditionType = PermanentGlobal

2. Gear Crit Rate

This is associated with:

Equipped Gear | Gloves / SS Gear / Necklace | Conditional DPS

The source shows multiple gear-based crit sources.

Important gear associations visible:

Eternal Gloves

From the gloves source, Eternal Gloves are heavily tied to crit.

Relevant effects:

- Excellent / purple: Attacks have 10% chance to crit.
- Legendary / red: Crit Rate +10%, Crit Damage +100%.

- AF3: Giantbane gives +3% Crit Rate when an elite or boss is defeated.

The crit source image also shows Eternal Gloves as:

0–40% Crit Rate, +10% with R5 collectible

This means Eternal Gloves crit value may depend on gear rarity, AF state, collectible support, or battle condition.

Leather Gloves

Leather Gloves are associated with:

- Crit Rate +5%
- Crit Damage +50%
- Crits inflict Bleed

This matters because it connects crit with the **Bleed / Laceration / debuff family**, not just raw crit.

SS / advanced gear sources

The image shows additional SS-style gear crit sources:

- 10% Crit Rate
- 20% Crit Rate
- AF3: +3% extra Crit Rate per boss / elite kill
- 30% Crit Rate
- V4: +20% Crit Rate during laser charge

Optimizer notes

Gear crit should **not** be treated as permanent global. It only counts if the item is equipped and the condition is active.

Database flags:

SourceCategory = Gear

Slot = Gloves / Necklace / SSGear

TargetScope = CurrentBuildOnly

ConditionType = Equipped / BattleConditional

DPSRelevant = High

Example row:

Eternal_Gloves | Gear | Gloves | Red | CritRate | +10% | CurrentBuild | Equipped | Yes | Also gives CritDamage +100%

Example row:

SS_Gear_Laser_Charge | Gear | UnknownSSSlot | V4 | CritRate | +20% | CurrentBuild | During laser charge | Yes | Temporary burst condition

3. Skill Crit Rate

This is associated with:

In-Battle Skills | Survivor Skill | Stance | Temporary DPS

The image shows skill-based crit rate sources:

- King-related skill source: Crit Rate +40%
- King-related evolved / upgraded source: Crit Rate +50%
- Yang in Palm stance: Crit Rate +20%

Optimizer notes

These should be stored as **in-run conditional sources**, not account stats.

Important: Yang's Palm stance crit should only count when the optimizer is modeling **Master Yang in Palm stance**. Do not apply it to every survivor.

Database flags:

SourceCategory = Skill

TargetScope = ActiveRunOnly

ConditionType = SkillSelected / StanceActive

DPSRelevant = High

Example row:

Yang_Palm_Stance | Skill | MasterYang | CritRate | +20% | ActiveSurvivorOnly | Palm stance active | Yes

4. Collectible Crit Rate

This is associated with:

Collectibles | Red-star thresholds | Global DPS

The source shows two collectible groups for Crit Rate:

Red collectible group

- 10 collectibles
- Each gives Crit Rate +10%
- Total shown: **100% global Crit Rate**

Yellow collectible group

- 11 collectibles
- Each gives Crit Rate +5%
- Total shown: **55% global Crit Rate**

Optimizer notes

This is extremely important later-game account scaling.

For now, because the image uses icons, the optimizer should store each collectible by icon/source until we map the exact collectible names from the collectible edition images.

Database flags:

SourceCategory = Collectible

TargetScope = Global

ConditionType = OwnedCollectibleStarThreshold

DPSRelevant = Very High

Example row:

Collectible_CR_Icon_01 | Collectible | UnknownNamePending | CritRate | +10% | Global |
Requires listed star threshold | Yes | Map exact name from Collectibles Edition source

5. Master Yang awakening Crit Rate

This is associated with:

Master Yang | Awakening Red Stars | Active Survivor DPS

The source shows Yang / panda crit rate from red awakening stars:

- 1 red star: Crit Rate +5%
- 4 red stars: additional +10%
- 6 red stars: additional +15%
- Total: **30% Crit Rate**

Optimizer notes

This should probably be treated as **Master Yang active-survivor value**, not global, unless another source confirms it is global.

Database flags:

SourceCategory = SurvivorAwakening
Survivor = MasterYang
StatType = CritRate
TargetScope = ActiveSurvivorLikely
ConditionType = AwakeningLevel
DPSRelevant = High

6. Talent Crit Rate

This is associated with:

Talents | Expose Weakness | Global Account Stat

The source shows:

- Talent: **Expose Weakness**
- Level 40
- Crit Rate +8%
- Marked as global

Optimizer notes

This is a clean global stat.

Database row:

Expose_Weakness_Lv40 | Talent | CritRate | +8% | Global | Permanent | Yes

7. Pet Crit Rate

This is associated with:

Pet Skill | Motivation | Pet / build-dependent DPS

The source shows:

- Pet skill: **Motivation**
- Crit Rate +3% / +4% / +5%

The value likely depends on the pet skill level / rarity.

Optimizer notes

This should only count when the pet has the skill equipped/active.

Database flags:

SourceCategory = PetSkill

PetSkill = Motivation

StatType = CritRate

Value = 3/4/5%

TargetScope = CurrentPetBuild

ConditionType = EquippedPetSkill

DPSRelevant = Medium/High

8. Donatello Crit Rate passive

This is associated with:

Donatello | Brotherhood | TMNT Passive Count | Global / Loadout Scaling

The source shows:

- Donatello 1 red star
- Crit Rate: 3% / 6% / 9%
- Text: **3% per equipped TMNT passive**

This connects directly to the previous SP survivor source.

Optimizer notes

This is one of the most important synergy stats because it scales based on how many TMNT/SP passives are equipped.

Database flags:

SourceCategory = SurvivorPassive

Survivor = Donatello

Passive = Brotherhood

StatType = CritRate

ScalingInput = EquippedTMNTPassiveCount

ValuePerPassive = +3%

MaxShown = +9%

ConditionType = DonatelloAwakeningUnlocked + TMNTPassivesEquipped

DPSRelevant = Very High

Example row:

Donatello_Brotherhood_CR | SurvivorPassive | Donatello | CritRate | +3% per equipped TMNT passive | LoadoutScaling | Requires Don A1 | Yes

9. Combat Harmony Crit Rate

This is associated with:

Combat Harmony | Star Level | Global Account Stat

The source shows:

- Combat Harmony level 1: Crit Rate +2%
- Level 3: additional +3%
- Level 6: additional +5%
- Level 9: additional +5%
- Level 12: additional +5%
- Level 15: additional +5%
- Level 18: additional +5%

Total shown:

30% global Crit Rate

Optimizer notes

This is global and should have high priority in the account stat model.

Database row:

Combat_Harmony_CR | CombatHarmony | CritRate | +30% total | Global | Based on harmony level | Yes

Crit Damage section

1. Default Crit Damage

The source shows:

Default Crit Damage = 100%

This should be stored as the baseline, not as an upgrade.

Important: do not accidentally double count this as a bonus source.

Database row:

Default_Crit_Damage | BaseStat | CritDamage | 100% | AllBuilds | Baseline | Yes | Do not treat as upgrade

2. Survivor level Crit Damage

This is associated with:

Survivor Level | Global Crit Damage | Account DPS

The source shows **60% total global Crit Damage** from survivor level bonuses.

Visible sources include:

- One survivor at level 40: Crit Damage +5%
- King level 120: Crit Damage +5%
- Tsukuyomi level 120: Crit Damage +5%
- Worm level 120: Crit Damage +5%

- Metallia / red-haired survivor level 120: Crit Damage +10%
- Catnips / cat-like survivor level 120: Crit Damage +10%
- SpongeBob level 40: Crit Damage +5%
- Squidward level 40: Crit Damage +5%
- Patrick level 40: Crit Damage +5%
- Sandy level 40: Crit Damage +5%

Optimizer notes

This is permanent account DPS scaling.

Database flags:

SourceCategory = SurvivorLevel
 StatType = CritDamage
 TargetScope = AllSurvivors
 ConditionType = PermanentGlobal
 DPSRelevant = Very High

3. Gear Crit Damage

This is associated with:

Equipped Gear | Gloves / SS gear / Necklace | Conditional DPS

The source shows several gear Crit Damage sources:

- V2 gear source: Crit Damage +36%
- V4 gear source: Crit Damage +60%
- Leather Gloves: Crit Damage +50%
- Eternal Gloves red: Crit Damage +100%
- Gear + collectible condition: Crit Damage +20%
- SS necklace / energy shield condition:
 - E5: 0–50%, listed as 10% per energy level
 - E1: if Crit Rate >100%, Crit Damage +30%
 - E5: if Crit Rate >150%, Crit Damage +100%

Optimizer notes

This is where crit math becomes conditional. The optimizer needs to know the player's **Crit Rate** before applying some Crit Damage bonuses.

Especially important:

If Crit Rate >100%, add +30% Crit Damage

If Crit Rate >150%, add +100% Crit Damage

That means Crit Rate can unlock more Crit Damage. This makes Crit Rate more valuable than it looks.

Database flags:

SourceCategory = Gear

StatType = CritDamage

ConditionType = Equipped / Threshold / EnergyLevel

DependsOn = CritRate / EnergyLevel / GearRarity

DPSRelevant = Very High

Example row:

SS_Necklace_CR_Threshold_E1 | Gear | Necklace | CritDamage | +30% | CurrentBuild |
Requires CritRate >100% | Yes

Example row:

SS_Necklace_CR_Threshold_E5 | Gear | Necklace | CritDamage | +100% | CurrentBuild |
Requires CritRate >150% | Yes

4. Skill Crit Damage

This is associated with:

Survivor Skill | Awakening Stars | Active Survivor DPS

The source shows:

Metallia / red-haired survivor skill

- 1 red star: Crit Damage +20%
- 3 red stars: additional +20%
- 5 red stars: additional +20%

Total shown:

+60% Crit Damage

King evolved skill

- King-related evolved skill: Crit Damage +25%

Optimizer notes

These are not global. They depend on active survivor and skill state.

Database flags:

SourceCategory = Skill

StatType = CritDamage

TargetScope = ActiveRunOnly

ConditionType = ActiveSurvivor / SkillUnlocked / AwakeningLevel

DPSRelevant = High

5. Collectible Crit Damage

This is associated with:

[Collectibles](#) | [Red-star thresholds](#) | [Global DPS](#)

The source shows two collectible groups:

Red collectible group

- 10 collectibles
- Each gives Crit Damage +10%
- Total: **100% global Crit Damage**

Yellow collectible group

- 13 collectibles
- Each gives Crit Damage +5%
- Total: **65% global Crit Damage**

Optimizer notes

This should be connected to the collectible database later. For now, store by icon ID and source image.

Database flags:

SourceCategory = Collectible

StatType = CritDamage

TargetScope = Global

ConditionType = OwnedCollectibleStarThreshold

DPSRelevant = Very High

6. Master Yang awakening Crit Damage

This is associated with:

Master Yang | Awakening Red Stars | Active Survivor DPS

The source shows:

- 1 red star: Crit Damage +5%
- 4 red stars: additional +10%
- 6 red stars: additional +25%

Total shown:

40% Crit Damage

Optimizer notes

Treat this as Yang-specific unless another source confirms it is global.

Database flags:

SourceCategory = SurvivorAwakening

Survivor = MasterYang

StatType = CritDamage

ValueTotal = +40%

TargetScope = ActiveSurvivorLikely

ConditionType = AwakeningLevel

DPSRelevant = High

7. Survivor Synergy Crit Damage

This is associated with:

Survivor Synergy | Global Account Stat | Crit Damage

The source shows:

- Synergy level 5: Crit Damage +5%
- Synergy level 10: additional +5%
- Synergy level 15: additional +10%
- Synergy level 20: additional +10%

Total:

30% global Crit Damage

Optimizer notes

This is clean global DPS.

Database row:

Survivor_Synergy_CD | SurvivorSynergy | CritDamage | +30% total | Global | Based on synergy level | Yes

8. Pet Crit Damage

This is associated with:

Pet | Rex | Inspiration | Active Pet DPS

The source shows two pet-related Crit Damage sources.

Rex

- Rex at 3 yellow stars: Crit Damage +5%
- Rex at red stars: Crit Damage +10%
- Note: **Rex must be active**

This means Rex is not just a passive account source. It only counts when Rex is deployed.

Inspiration

- Pet skill: **Inspiration**
- Crit Damage +6% / +10%

Optimizer notes

Pet crit damage should only be counted if the correct pet/skill is active.

Database flags:

SourceCategory = Pet

Pet = Rex

StatType = CritDamage

ConditionType = ActivePetRequired

DPSRelevant = Medium/High

Example row:

Rex_Active_CD | Pet | Rex | CritDamage | +5%/+10% | CurrentPetBuild | Rex must be active | Yes

Example row:

PetSkill_Inspiration | PetSkill | Inspiration | CritDamage | +6%/+10% | CurrentPetBuild | Skill equipped | Yes

9. Donatello Crit Damage passive

This is associated with:

[Donatello](#) | [Brotherhood](#) | [TMNT Passive Count](#) | [Loadout Scaling](#)

The source shows:

- Donatello 2 red stars
- Crit Damage: 3% / 6% / 9%
- Text: **3% per equipped TMNT passive**

This connects with Donatello's Brotherhood passive from the SP survivor source.

Database flags:

SourceCategory = SurvivorPassive

Survivor = Donatello

Passive = Brotherhood
StatType = CritDamage
ScalingInput = EquippedTMNTPassiveCount
ValuePerPassive = +3%
MaxShown = +9%
ConditionType = DonatelloAwakeningUnlocked + TMNTPassivesEquipped
DPSRelevant = Very High

Most important optimizer takeaways

1. Crit Rate is not just a damage stat — it unlocks Crit Damage bonuses

Because some gear gives extra Crit Damage only when Crit Rate is above a threshold:

Crit Rate >100% = extra Crit Damage

Crit Rate >150% = much bigger extra Crit Damage

So the calculator should give extra value to Crit Rate when it helps cross those breakpoints.

2. Global crit sources are the best long-term investments

Highest-value global sources in this image:

Survivor Levels
Collectibles
Combat Harmony
Talents
Survivor Synergy
TMNT 5-star bonuses

These help every build and every mode.

3. Yang, King, Metallia, Donatello, and pets should be conditional

These should not be blindly added to all builds.

Use conditions like:

ActiveSurvivor = MasterYang

ActiveSurvivor = King

ActiveSurvivor = Metallia

DonatelloPassiveUnlocked = True

EquippedTMNTPassiveCount = 0/1/2/3

ActivePet = Rex

PetSkill = Inspiration or Motivation

Clean database format for this data type

Use this table structure:

CritSourceID

StatType

SourceCategory

SourceName

AssociatedCharacter

AssociatedItem

AssociatedPet

AssociatedSkill

UnlockRequirement

Value

ValueType

TargetScope

ConditionType

IsGlobal

IsConditional

DPSRelevant

StackingGroup

Notes

SourceImage

SourceDate

Example rows:

TMNT_Raphael_5Star_CR | CritRate | SurvivorStar | Raphael | None | None | None | 5 yellow stars | 8% | FlatPercent | AllSurvivors | Permanent | Yes | No | VeryHigh | GlobalCritRate | Special event survivor | Crit_Rate_And_Damage_2025-03-09

Donatello_Brotherhood_CR | CritRate | SurvivorPassive | Donatello | None | None | Brotherhood
| Don A1 + equipped TMNT passives | 3% per passive | ScalingPercent | CurrentLoadout |
PassiveCount | Maybe | Yes | VeryHigh | LoadoutCritRate | Max shown 9% |
Crit_Rate_And_Damage_2025-03-09
SS_Necklace_CR_Threshold_CD | CritDamage | Gear | None | SS Necklace | None | None |
Crit Rate >150% | 100% | FlatPercent | CurrentBuild | Threshold | No | Yes | VeryHigh |
ConditionalCritDamage | Requires crit breakpoint | Crit_Rate_And_Damage_2025-03-09
Rex_Active_CD | CritDamage | Pet | None | None | Rex | None | Rex active | 5%/10% |
FlatPercent | CurrentPetBuild | ActivePet | No | Yes | MediumHigh | PetCritDamage | Rex must
be active | Crit_Rate_And_Damage_2025-03-09

Formula logic for the optimizer

Store the pieces separately first:

TotalCritRate =
GlobalCritRate
+ GearCritRate_Equipped
+ SkillCritRate_Active
+ PetCritRate_Active
+ SurvivorPassiveCritRate_Active
+ ConditionalCritRate_Active
TotalCritDamage =
DefaultCritDamage
+ GlobalCritDamage
+ GearCritDamage_Equipped
+ SkillCritDamage_Active
+ PetCritDamage_Active
+ SurvivorPassiveCritDamage_Active
+ ConditionalCritDamage_Active

Then calculate DPS after checking conditions:

If TotalCritRate > 100:
 apply CritDamage threshold bonuses that require CR > 100

If TotalCritRate > 150:
 apply CritDamage threshold bonuses that require CR > 150

Do **not** hardcode the final damage formula yet until we verify Survivor.io's exact crit math. But for ranking choices, this source clearly says Crit Rate and Crit Damage are major DPS multipliers and should be weighted very heavily.

Data type 3: Pets, Pet Skills, Pet Awakening

Sources used here:

Pets_2024-11-28
Pet_Skills_2024-09-09
Pet_Awakening_Guide_2024-10-25
Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04
Pet_Awakening_Guide_-_Murica_2024-10-10
Pet_Awakening_Guide_-_Croaky_2024-10-11
Pet_Awakening_Guide_-_Rex_2024-10-11

This data type belongs in the optimizer as:

Pet | Pet Skill | Pet Awakening | Assist Skill | Pet Toy | Pet Affection | Pet DPS | Owner DPS Buff
| Pet Crit | Pet Cooldown | Pet Inherited ATK | Pet Utility

For your calculator, pets matter in **two different ways**:

1. **Direct pet damage** — Murica/Eagle whirlwind, Crab bullets, Robot lasers, etc.
2. **Owner DPS buffs** — Crit Rate, Crit Damage, Skill Damage, cooldown, pet skill damage, inherited ATK.

HP, damage reduction, revival time, and invincibility should be stored, but mostly marked as **ignore for DPS** unless later you make a survival-mode version.

Main pet structure

Every pet has:

Primary Attack
Fixed pet skill pool
Awakening/star progression
Assist skill compatibility
Disposition
Toy/affection bonus

The optimizer should not treat all pets equally. A pet is valuable based on:

PetBaseAttack
PetAttackCooldown
PetSkillDamage
PetCooldownReduction
PetCritRate
PetCritDamage
PetInheritedATK
OwnerCritRateBuff
OwnerCritDamageBuff
OwnerSkillDamageBuff
AssistSkillSetup
AwakeningLevel

Pet list from **Pets_2024-11-28**

1. Frog / Croaky

Main association

Frog | Croaky | Corrosive Saliva | Poison/DOT style damage | Owner Skill Damage | Crit Rate | Crit Damage

Primary attack

Frog spits corrosive saliva that deals:

200% ATK damage every 1 second for 2 seconds
Cooldown: 1 second

This is a damage-over-time style attack.

Built-in skill themes

Frog/Croaky is associated with:

Owner Skill Damage
Pet inherited HP
Battle EXP gain
Owner Crit Rate
Owner Crit Damage

Visible pet effects include:

Owner Skill Damage +3%

Pet inherited HP +10%

Owner Battle EXP gain +7.5%

Owner Crit Rate +4%

Owner Crit Damage +10%

Optimizer notes

Croaky is not just a pet attack pet. It also has **owner DPS buff value** because it can connect to Skill Damage, Crit Rate, and Crit Damage.

The pet awakening guide says **Rex or Croaky should be taken to R5 first**, because both are close in total damage.

Database flags:

Pet = Croaky

PrimaryAttack = Corrosive Saliva

DamageType = DOT / Poison-like

OwnerSkillDamageSource = Yes

OwnerCritRateSource = Yes

OwnerCritDamageSource = Yes

PetDirectDamage = Medium/High

DPSRelevant = High

2. Fish

Main association

Fish | Explosive Bubbles | Explosion Damage | Battle EXP | Time Stop

Primary attack

Fish spits explosive bubbles:

Bubble hit damage: 200% ATK

Explosion damage: 200% ATK

Cooldown: 1 second

Built-in skill themes

Fish is associated with:

Battle EXP gain
Collision damage reduction
Shot damage reduction
Pet size/range
Time Stop

The red skill shown is:

When owner's HP is below 30%, stops time and freezes all enemies for 5 seconds.
During this, player and pet are invincible.
Cooldown: 60 seconds.

Optimizer notes

Fish has some damage from explosion, but a lot of its value is **utility/survival**, not pure DPS.

Database flags:

Pet = Fish
PrimaryAttack = Explosive Bubbles
DamageType = Explosion / AoE
Utility = Time Stop
DPSRelevant = Medium
IgnoreForDPS = Time Stop, damage reduction, survival effects

3. Dog / Rex

Main association

Dog | Rex | Yell Attack | Owner Crit Rate | Owner Crit Damage | Damage Transfer | Pet HP Absorb

Primary attack

Dog yells at enemies with loud noises:

250% ATK damage

Cooldown: 1 second

Built-in skill themes

Dog/Rex is associated with:

Owner Crit Rate
Owner Crit Damage
Pet movement speed
Pet HP absorb
Damage transfer

Visible pet effects include:

Owner Crit Rate +3%
Owner Crit Damage +6%
Pet movement speed +75%
Pet HP absorb from attacks +2%
Takes 30% damage in place of owner

Rex awakening guide

The Rex guide recommends assist skills:

Raid*
Strong
Anything

Disposition:

Hot-blooded
Battler

Rex shares the same key owner-buff skills as Croaky:

Battle Lust
Diligence
Encouragement
Inspiration
Motivation
Vampirism

Optimizer notes

Rex matters because it can support **owner Crit Rate and Crit Damage**, and the awakening cost guide specifically says Rex or Croaky should be pushed to R5 first.

Database flags:

Pet = Rex
PrimaryAttack = Yell
OwnerCritRateSource = Yes
OwnerCritDamageSource = Yes
SurvivalUtility = Damage transfer
DPSRelevant = High

4. Turtle

Main association

Turtle | Shockwave | Damage Reduction | Revival | Tank Pet

Primary attack

Turtle leaps up and smashes the ground:

Shockwave damage: 300% ATK
Cooldown: 1 second

Built-in skill themes

Turtle is associated with:

Shot damage reduction
Collision damage reduction
Pet revival time reduction
Pet inherited HP
Pet damage taken reduction

Visible effects include:

Owner Shot Damage Reduction +10%
Owner Collision Damage Reduction +10%
Pet revival time -30%
Pet inherited HP +15%

Pet damage taken -60%

Optimizer notes

Turtle is mostly a **survival/tank pet**. For your damage-only calculator, most of this gets stored but not scored.

Database flags:

Pet = Turtle

PrimaryAttack = Shockwave

DamageType = AoE

TankUtility = High

DPSRelevant = Low/Medium

IgnoreForDPS = Most defensive effects

5. Eagle / Murica

Main association

Eagle | Murica | Whirlwind | Penetrating Damage | Cooldown Reduction | Crush | Pet Skill Damage

Primary attack

Eagle flaps wings to create a whirlwind:

350% ATK damage

Whirlwind penetrates all enemies

Cooldown: 1 second

This is why Murica is so strong early/mid game. Penetrating attacks are very valuable when they hit multiple enemies or bosses consistently.

Built-in skill themes

Murica is associated with:

Pet size/range

Pet ATK damage

Pet cooldown reduction
Pet movement speed
Crush / 10x damage chance

Visible effects include:

Pet becomes bigger
Pet ATK and skill range +15%
Pet ATK damage +15%
Pet ATK and skill cooldown -20%
Pet movement speed +75%
Attacks have 5% chance to deal 10x damage to enemies below 50% HP

Murica awakening guide

The guide recommends assist skills:

Raid*
Powerful
Boost
Penetration

Disposition:

Shrewd
Hot-blooded
Battler

Important note from the source:

Replace Motivation with Inspiration if high Crit Rate %
After Y5, look at switching to Rex or Croaky.
Otherwise, use the Pet Optimizer.

Optimizer notes

Murica should be modeled as a **direct pet DPS pet**.

Its strongest associations are:

Whirlwind
Penetration
Cooldown reduction
Skill damage
Crush

Pet ATK damage

Database flags:

Pet = Murica

PrimaryAttack = Whirlwind

DamageType = Penetrating

PetCooldownReduction = Yes

PetSkillDamage = Yes

Crush = Yes

DirectPetDPS = Very High

DPSRelevant = Very High

6. Cat

Main association

Cat | Claw Attack | Pet Crit Rate | Pet Cooldown | Nine Lives

Primary attack

Cat scratches enemies with sharp claws:

200% ATK damage

Cooldown: 0.5 seconds

This means it attacks faster than most pets.

Built-in skill themes

Cat is associated with:

Pet ATK and skill Crit Rate

Pet inherited HP

Pet cooldown reduction

Pet ATK damage

Fatal-damage protection

Visible effects include:

Pet ATK and skill Crit Rate +15%

Pet inherited HP +10%
Pet ATK and skill cooldown -20%
Pet ATK damage +20%
When player and pet take fatal damage, nullify that attack and heal allies for 50% HP
Cooldown: 100 seconds

Optimizer notes

Cat has real DPS value because of **fast attack speed, pet crit, pet cooldown, and pet ATK damage**. The fatal-damage protection is survival, so store it but ignore for DPS.

Database flags:

Pet = Cat
PrimaryAttack = Claws
AttackCooldown = 0.5s
PetCritRateSource = Yes
PetCooldownReduction = Yes
PetATKDamageSource = Yes
DPSRelevant = High

7. Crab

Main association

Crab | Water Bullets | Pet ATK Damage | Pet Crit Rate | Pet Inherited ATK | Pet Cooldown

Primary attack

Crab fires powerful water bullets:

500% ATK damage
Cooldown: 1 second

Built-in skill themes

Crab is associated with:

Pet HP absorb
Pet ATK damage

Pet ATK and skill Crit Rate
Pet inherited ATK
Pet cooldown reduction

Visible effects include:

Pet HP absorb from attacks +1%
Pet ATK damage +15%
Pet ATK and skill Crit Rate +20%
Pet inherited ATK +15%
Pet ATK and skill cooldown -30%

Optimizer notes

Crab is a clean **pet DPS scaling pet**. Unlike Turtle/Fish, most of Crab's good stats are directly damage-related.

Database flags:

Pet = Crab
PrimaryAttack = Water Bullets
PrimaryDamage = 500% ATK
PetCritRateSource = Yes
PetCooldownReduction = Yes
PetInheritedATKSource = Yes
PetATKDamageSource = Yes
DPSRelevant = High/Very High

Since you have cared about Crab before, this source confirms Crab is not random trash. It has a real DPS identity, especially through cooldown, crit rate, inherited ATK, and 500% primary attack.

8. Robot

Main association

Robot | Burning Lasers | Satellite Cannon | Pet Cooldown | Inherited ATK

Primary attack

Robot fires 3 burning lasers:

3 lasers

Each laser deals 400% ATK damage

Cooldown: 1 second

If all lasers connect, this can be very high theoretical damage.

Built-in skill themes

Robot is associated with:

Pet cooldown reduction

Pet revival time

Pet damage reduction

Pet inherited ATK

Satellite Cannon

Visible effects include:

Pet ATK and skill cooldown -15%

Pet revival time -20%

Pet damage taken -40%

Pet inherited ATK +15%

Satellite blast deals 1000% ATK damage

Cooldown: 5 seconds

+250% damage when owner levels up

Optimizer notes

Robot has high theoretical DPS because of lasers and Satellite Cannon, but the calculator needs to eventually model **hit consistency**. If the lasers miss or split badly, the real value drops.

Database flags:

Pet = Robot

PrimaryAttack = Burning Lasers

PrimaryDamage = 3 x 400% ATK

SpecialSkill = Satellite Cannon

SpecialDamage = 1000% ATK

Cooldown = 5s

LevelUpScaling = +250% damage when owner levels up

DPSRelevant = High but hit-dependent

Pet skill registry

Source: [Pet_Skills_2024-09-09](#) and [Pet_Awakening_Guide_2024-10-25](#)

This is the skill dictionary. These skills should be stored separately from pets because different pets unlock different skills at different rarity/star levels.

Owner / Survivor buff skills

These affect the player, not only the pet.

Barrier

Increases player's Range Damage Reduction by 10/15%

DPSRelevant = No

Diligence

Increases player's Battle EXP gain by 5/7.5/10%

DPSRelevant = Indirect only

Encouragement

Increases player's Skill Damage by 3/5%

DPSRelevant = Yes

Guard

Increases player's Collision Damage Reduction by 10%

DPSRelevant = No

Inspiration

Increases player's Crit Damage by 6/10%

DPSRelevant = Yes

Motivation

Increases player's Crit Rate by 3/4/5%

DPSRelevant = Yes

Optimizer notes

For your damage-only calculator, the big owner-buff pet skills are:

Encouragement = Skill Damage

Inspiration = Crit Damage

Motivation = Crit Rate

Diligence should be stored because faster leveling can matter in chapters, but it is not a direct boss DPS stat.

Pet DPS skills

These affect the pet's own damage.

Crush

Attacks have 5% chance to deal 10x damage

DPSRelevant = Very High

Battle Lust

Reduces pet cooldowns by 15/20/30%

DPSRelevant = Very High

Frenzy

Increases pet Crit Rate by 15/20/60%

DPSRelevant = Very High

Gargantua

Pet becomes bigger and range increases by 15/20%

DPSRelevant = Medium/High because it improves hit coverage

High Windchaser

Pet movement speed +75%

DPSRelevant = Low/Medium, mostly hit consistency

Painstrike

Increases pet Crit Damage by 60%

DPSRelevant = High

Sharp Claws

Increases pet Skill Damage by 15/20/60%

DPSRelevant = Very High

Tough Skin

Reduces pet damage taken by 40/60%

DPSRelevant = No

United Effort

Increases pet inherited HP by 10/15/35%

DPSRelevant = No

United Stand

Increases pet inherited ATK by 15/35%

DPSRelevant = High

Unyielding Will

Reduces pet revival time by 20/30/40%

DPSRelevant = Mostly no

Utmost Loyalty

Pet takes 30% damage in place of player
DPSRelevant = No
Vampirism
Increases pet HP absorb from attacks by 1/2%
DPSRelevant = No, unless survival model is added

Misc pet skills

Nine Lives
When player and pet take fatal damage, nullifies that attack and heals allies for 50% HP.
Cooldown: 100 seconds.
DPSRelevant = No
Satellite Cannon
Guides satellite to blast a shot at a certain area.
Damage: 1000% ATK.
Cooldown: 5 seconds.
+250% damage when owner levels up.
DPSRelevant = Yes
Time Stop
When player's HP is below 30%, stops time and freezes enemies for 5 seconds.
Player and pet are invincible during it.
Cooldown: 60 seconds.
DPSRelevant = No, but useful utility

Assist skills

Assist skills are separate from the deployed pet's normal skill set. These should be modeled as:

AssistPetSkill | AssistPet | DeployedPetBuff | InBattleScaling

The assist skills shown:

Boost
For every 5 owner levels, pet Range increases by 5/6/7.5/10%.
DPSRelevant = Medium/High
Durable
For every 5 owner levels, pet shot Duration increases by 8/10/12/15%.
DPSRelevant = Medium/High if pet projectile duration matters
Penetration
For every 10/8/6/5 owner levels, penetrated units of pet shot +1.

DPSRelevant = High for projectile/whirlwind pets

Powerful

For every 5 owner levels, pet Skill Damage increases by 5/6/7.5/10%.

DPSRelevant = High

Strong

For every 5 owner levels, pet Max HP increases by 10/12/15/20%.

DPSRelevant = No

Raid

For every 5 owner levels, pet Skill Damage increases by 6/7.5/10%.

Bonus trait: deployed pet cooldowns reduced by 10/15/20%.

DPSRelevant = Very High

Important note:

Raid is marked as Gary exclusive.

That means the optimizer needs an availability check:

If GaryAvailable = false:

RaidAssistSkill = unavailable

Recommended pet awakening setups

Murica

Source: [Pet_Awakening_Guide_-_Murica_2024-10-10](#)

Murica recommended assist skills:

Raid*

Powerful

Boost

Penetration

Disposition:

Shrewd

Hot-blooded

Battler

Important source note:

Replace Motivation with Inspiration if high Crit Rate %
After Y5, look at making the switch to Rex or Croaky.
Otherwise, use the Pet Optimizer.

Optimizer interpretation

Murica is great when the build needs:

Direct pet DPS
Whirlwind damage
Cooldown reduction
Skill damage scaling
Penetration/range scaling

But once your account has high crit and stronger pet awakenings, the guide says to consider Rex or Croaky after Y5.

Croaky

Source: [Pet_Awakening_Guide_-_Croaky_2024-10-11](#)

Croaky recommended assist skills:

Boost
Raid*
Strong

Disposition:

Shrewd
Hot-blooded
Battler

Important note:

Use a lower tier of Battle Lust for every tier of Raid used.

Optimizer interpretation

Croaky is strong because it mixes:

Owner Skill Damage

Owner Crit Rate
Owner Crit Damage
Pet damage scaling
Assist scaling

Croaky should be scored as a strong long-term pet, not just as an early pet.

Rex

Source: [Pet_Awakening_Guide_-_Rex_2024-10-11](#)

Rex recommended assist skills:

Raid*
Strong
Anything

Disposition:

Hot-blooded
Battler

Optimizer interpretation

Rex is valuable because it has:

Owner Crit Rate
Owner Crit Damage
Damage transfer utility
Pet HP absorb

The pet cost guide also says Rex or Croaky should be taken to R5 first because they are close in total damage.

Pet awakening costs and affection bonuses

Source: [Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04](#)

This is a cost/progression table. It matters because the optimizer needs to decide where awakening crystals should go.

Total awakening crystals

Total crystals to full pet awakening path shown: 350

Awakening crystal costs by stage

Y1: +5 crystals
Y2: +5 crystals
Y3: +10 crystals
Y4: +20 crystals
Y5: +30 crystals
R1: +40 crystals
R2: +60 crystals
R3: +60 crystals
R4: +60 crystals
R5: +60 crystals
Total: 350 crystals

Deploy count shown

Y1: 3 deployed pets
Y2: 3 deployed pets
Y3: 5 deployed pets
Y4: 5 deployed pets
Y5: 7 deployed pets
R1: 7 deployed pets
R2: 9 deployed pets
R3: 9 deployed pets
R4: 9 deployed pets
R5: 15 deployed pets

Affection level caps

Base: Level 10
Y1: Level 20
Y2: Level 30
Y3: Level 40
Y4: Level 50
Y5: Level 60
R1: Level 70
R2: Level 80

R3: Level 90
R4: Level 105
R5: Level 130

Player ATK from affection

The table gives player ATK per affection level and max ATK.

Important max ATK values:

Base: 300 max ATK
Y1: 900 max ATK
Y2: 1800 max ATK
Y3: 2700 max ATK
Y4: 3600 max ATK
Y5: 4800 max ATK
R1: 6000 max ATK
R2: 7500 max ATK
R3: 9000 max ATK
R4: 11.3k max ATK
R5: 15k max ATK

Optimizer notes

This is very important because pet awakening is not only pet damage. It gives **player ATK** through affection.

For the calculator:

PetAffectionATK = Global/player ATK source
PetAwakeningCrystalCost = investment cost
PetAwakeningStage = unlock requirement

HP values should be stored but not scored for your DPS optimizer.

Preferred pet toys

The source groups toys by pet pair.

Dog + Cat
Robot + Frog

Fish + Eagle
Turtle + Crab

Toy stat values shown:

Blue toy: 3 ATK, 12 HP
Purple toy: 15 ATK, 60 HP
Yellow toy: 45 ATK, 180 HP

Important note:

You must level up affection before gaining the benefits.

Optimizer notes

Toy stats are small individually but permanent. Since your optimizer cares about damage:

ToyATK = count
ToyHP = store but ignore

Pet awakening investment advice from the source

The source gives a clear priority rule:

1. Take Rex or Croaky to R5 first. Both are very close in total damage.
2. After that, take each pet to Y5 with a balanced approach to get the most ATK per crystal.
Example: all pets to Y2, then all to Y3, etc.
3. Finally, take one pet from each gift type to R5.

It also says:

Since Y5 to R5 takes 280 crystals, the average for all 5 levels is about 36.5 crystals.
It is not worth rushing R1-R3 at worse rates when cheaper Y-level sources still exist.

Optimizer interpretation

This means the optimizer should not blindly push one pet from Y5 to R1/R2/R3 if the player has cheaper account-wide ATK available from other pets.

The calculator needs a value function like:

ATK gained per awakening crystal
DPS gained per awakening crystal
Unlock value from new skill slot
Direct pet damage gained
Owner DPS buff gained

Biggest DPS-relevant pet stats

For the optimizer, the pet system should prioritize:

Owner Crit Rate
Owner Crit Damage
Owner Skill Damage
Pet Skill Damage
Pet ATK Damage
Pet Crit Rate
Pet Crit Damage
Pet Cooldown Reduction
Pet Inherited ATK
Pet Range / Penetration
Pet Primary Attack Damage
Pet Affection ATK

Ignore or down-rank:

Pet HP
Owner HP
Damage reduction
Revival time
Invincibility
Damage transfer
Time stop

Store them anyway, but mark:

DPSRelevant = No

Clean database format for this data type

Use these tables.

Pet base table

PetID
PetName
BaseAnimal
PrimaryAttackName
PrimaryAttackDescription
PrimaryDamagePercent
PrimaryCooldown
DamageType
HitType
DPSRelevant
Notes
SourceImage
SourceDate

Example:

PET_MURICA | Murica | Eagle | Whirlwind | Creates penetrating whirlwind | 350% ATK | 1s |
Penetrating | Multi-hit/line | Yes | Strong direct pet DPS | Pets_2024-11-28

Example:

PET_CRAB | Crab | Crab | Water Bullets | Fires powerful water bullets | 500% ATK | 1s |
Projectile | Bullet | Yes | Strong pet DPS scaling from crit/CD/ATK | Pets_2024-11-28

Pet skill table

PetSkillID
SkillName
SkillCategory
AffectedTarget
AffectedStat
ValuePurple
ValueYellow
ValueRed
Condition
DPSRelevant
Notes
SourceImage
SourceDate

Example:

PETSKILL_MOTIVATION | Motivation | SurvivorBuff | Owner | CritRate | 3% | 4% | 5% | Pet skill unlocked/equipped | Yes | Owner DPS crit source | Pet_Skills_2024-09-09

Example:

PETSKILL_BATTLE_LUST | Battle Lust | PetBuff | Pet | CooldownReduction | 15% | 20% | 30% | Pet skill unlocked/equipped | Yes | Very high pet DPS scaling | Pet_Skills_2024-09-09

Example:

PETSKILL_CRUSH | Crush | PetBuff | Pet | DamageProc | None | None | 5% chance 10x damage | Pet attacks | Yes | Strong burst pet DPS | Pet_Skills_2024-09-09

Pet awakening cost table

AwakeningStage
CrystalCost
TotalCrystalCost
DeployCountRequired
SacrificeRequirement
AffectionLevelCap
PlayerATKPerLevel
MaxPlayerATK
HPPerLevel
MaxHP
DPSRelevant
SourceImage
SourceDate

Example:

Y5 | 30 | 70 total to Y5 | 7 | Red pet sacrifice | 60 | +120 | 4800 | +480 | 28.8k | Yes for ATK, no for HP | Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04

Example:

R5 | 60 | 350 total | 15 | Red pet sacrifice | 130 | +150 | 15k | +600 | 78k | Yes for ATK, no for HP | Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04

Pet recommendation table

PetID
RecommendedStage

RecommendedAssist1
RecommendedAssist2
RecommendedAssist3
RecommendedAssist4
PreferredDisposition1
PreferredDisposition2
PreferredDisposition3
SpecialNote
DPSPriority
SourceImage
SourceDate

Example:

PET_MURICA | Y5+ | Raid | Powerful | Boost | Penetration | Shrewd | Hot-blooded | Battler |
Replace Motivation with Inspiration if high Crit Rate; after Y5 consider Rex/Croaky | Very High
early/mid | Pet_Awakening_Guide_-_Murica_2024-10-10

Example:

PET_CROAKY | R5 target | Boost | Raid | Strong | None | Shrewd | Hot-blooded | Battler | Use
lower tier Battle Lust for each Raid tier used | Very High long-term |
Pet_Awakening_Guide_-_Croaky_2024-10-11

Example:

PET_REX | R5 target | Raid | Strong | Anything | None | Hot-blooded | Battler | None | Rex or
Croaky to R5 first; close total damage | Very High long-term |
Pet_Awakening_Guide_-_Rex_2024-10-11

Formula logic for pet DPS

Keep pet DPS separate first:

PetDPS =
PetBaseAttackDamage
* PetAttackFrequency
* PetATKMultiplier
* PetSkillDamageMultiplier
* PetCritMultiplier
* PetCooldownMultiplier
* HitConsistencyMultiplier

Then owner DPS buffs from pets:

OwnerDPSFromPetBuffs =
OwnerCritRateFromPet
+ OwnerCritDamageFromPet
+ OwnerSkillDamageFromPet
+ PlayerATKFromPetAffection

Do not mix these too early. A pet can be good because it does direct damage, or because it buffs the owner.

Important conclusion for the optimizer

The pet system is not “just pick Murica forever.”

The source data says:

Murica = strong direct pet DPS, especially through whirlwind, cooldown, Crush, and assist scaling.

Croaky = strong long-term owner/pet DPS mix.

Rex = strong long-term owner crit support and total damage contender.

Crab = real pet DPS scaling through 500% attack, crit rate, cooldown, and inherited ATK.

Robot = high theoretical laser/satellite damage but hit consistency matters.

Turtle/Fish = more utility/survival than damage.

For your damage-only optimizer, the highest-value pet stats are:

Battle Lust
Raid
Powerful
Sharp Claws
Frenzy
Crush
Motivation
Inspiration
Encouragement
United Stand
Pet Affection ATK

Data type 4: SS Gear / Relic Core Gear Pathing

Source: [SS_Gear_2025-02-06](#)

Game version shown: **3.6**

Updated: **2025-02-06**

This image is not just “equipment.” It is a full **SS gear upgrade map** showing:

- SS Gear Slot
- Base rarity effects
- Eternal forge path
- Void forge path
- Chaos forge path
- Chaos Fusion Power
- Collectible set bonuses tied to SS gear
- Relic/core/fodder cost rows
- Damage/debuff/crit/shield associations

For the optimizer, this should become:

SSGear | Slot | ForgePath | ForgeLevel | EffectName | StatAffected | Condition | DPSRelevant | Cost | AssociatedDebuff | AssociatedCollectibleSet

Important note: some official item names are not written clearly in the table, so I’m labeling them by **slot + effect name** when needed. That is safer than inventing a wrong item name.

SS gear slots shown

The image has 6 SS gear columns:

- SS Weapon
- SS Necklace
- SS Gloves
- SS Armor
- SS Belt
- SS Boots

DPS value by slot

For your damage-only optimizer:

SS Weapon = Very High DPS

SS Gloves = Very High DPS

SS Belt = Very High DPS

SS Armor = High DPS

SS Boots = High but condition-based DPS

SS Necklace = Medium/High, mostly skill/shield conditional

HP effects should be stored but mostly ignored for DPS.

1. SS Weapon

Main association

SS Weapon | Starforged Havoc | Havoc Nova | Eternity Ray | Void Fissure | Void Power | Energy Overload | Chaos Meteor | Chilled/Weakened damage

This is one of the most important DPS systems because it combines:

Skill damage

Attack speed/interval changes

Void Fissure damage taken debuff

Energy Overload stacking

Chaos path max-level increases

Chilled target damage

Weakened target damage

Boss lock-on behavior

Base / rarity effects

ATK +15%

Starforge:

Triggers Energy Overload 1 time for every 2 Eternity Ray attacks.

Increases own skill damage by 30% for 10 seconds.

Max 2 stacks.

Starforged Havoc:

Higher thrust damage with lower HP.

ATK +25%

Havoc Nova:

Releases Void power during a thrust.

Targets hit by Void Fissure take an additional 10% damage.

Havoc Nova:

Outermost rim of Eternity Ray has suppression force.

Eternal forge path

E1:

Energy Overload count also recorded when Eternity Ray retracts.

Starforge and Havoc start at Lv.3.

E2:

ATK +10%

E3:

Super Star:

Each Energy Overload stack's damage increased to 35%.

E4:

Final ATK +4000

E5:

Jade Cluster:

Energy Overload stacks up to 3 times.

Void forge path

V1:

ATK +10%

V2:

Disordered Death:

Void power damage +80%.

Targets hit by Void Fissure take an additional 20% damage.

V3:

Final ATK +4000

V4:

Blink:

Thrusts performed at low HP reduce base attack intervals by up to 1 second.

V5:

Final ATK +8000

Chaos forge path

This source shows Chaos path only under the weapon.

C1:

Max Level +5

C2:

Meteorite:

When Void Power is fired, additional homing Chaos Meteor fires are launched.

Inflicts 10% Vulnerability on hit for 10 seconds.

Does not stack.

C3:

Max Level +5

C4:

Damage to Weakened targets +20%

C5:

Max Level +5

C6:

Shatterstar:

When Chaos Meteorite hits enemies, Chaos Shatterstar is triggered.

C7:

Max Level +5

C8:

Damage to Chilled targets +30%

C9:

Max Level +10

C10:

Warp:

After Void Power hits a boss, it locks onto the boss aura and homes back in after flying out to deal damage again.

Void Power base damage -30%.

Chaos Fusion Power

3x:

Eternally Immortal:

Thrusts no longer depend on HP.

All effects trigger at full HP.

Targets hit by Void Fissure take +10% additional damage.

6x:

Infinity Void:

Void Power additionally gains 30% of Overload's damage bonus.

Targets hit by Void Fissure take +20% additional damage.

9x:

Chaos Wind:

Removes Eternity Ray and fuses its energy into Void Power, transforming it into Chaos Wind.

Post-fusion Chaos Wind triggers Energy Overload.

Each Overload stack's damage boost effect +25%.

Optimizer notes

This is a **core DPS item**. The calculator should track:

EnergyOverloadStacks
SkillDamageBonus
VoidFissureDamageTakenBonus
VoidPowerDamage
ChaosMeteorVulnerability
DamageToWeakened
DamageToChilled
BossLockOnBehavior
AttackIntervalReduction
MaxSkillLevelIncrease

Database flags:

Slot = Weapon

DPSRelevant = Very High

Debuffs = Vulnerability, Void Fissure damage taken, Weakened synergy, Chilled synergy

ModeValue = Very High for boss modes and general DPS

2. SS Necklace

Main association

SS Necklace | Skill Choice | Energy Shield | Clarity | Skill Damage | Reverse Scale | Shield HP

This is a mixed item. A lot of it is shield/HP, but the useful DPS parts are:

Extra skill choice

Skill damage

Shield-based skill damage

Clarity upgrade

Reverse Scale

Base / rarity effects

Skill choice chances when match starts +1

HP +15%

Adversity:

Gain an energy shield with 30% of Max HP when dying.

Once per battle.

Resets on revival.

HP +25%

Clarity:

Skill damage +20%.

Effect fades on death.

Preservation:

HP recovery through supplies also restores 10% of energy shield.

Shield recovery cannot exceed max HP.

Max 100% HP.

Eternal forge path

E1:

Halberd Shield:

Shield damage +10%.

Damage up when shield is active.

Taking damage lasts 10 seconds.

Energy shield slowly restores 2.5% HP every second.

Cannot exceed max shield HP.

E2:

HP +10%

E3:

Shield damage +10%.

Violent Backlash:

When own HP is higher than 10%, own HP is deducted first before shield HP when taking damage.

E4:

Final HP +16000

E5:

Omnicombat:

The higher the energy shield's HP, the higher the damage.

For every 25% shield HP, shield damage +5%.

Void forge path

V1:

HP +10%

V2:

Reverse Scale:

When shield HP is higher than own HP, skill damage +10%.

V3:

Final HP +16000

V4:

Clarity increases skill damage by +15%.

V5:

Final HP +32000

Optimizer notes

The necklace is not a simple raw DPS piece. It needs condition checks.

Track:

ExtraSkillChoice
SkillDamage
EnergyShieldPercent
ShieldHigherThanHP
DeathReset
ClarityActive
ShieldDamageBoost

Database flags:

Slot = Necklace
DPSRelevant = Medium/High
ConditionType = ShieldBased / DeathSensitive / SkillChoice
IgnoreForDPS = HP-only lines

3. SS Gloves

Main association

SS Gloves | Weaken | Damage to Weakened | Red Star passive upgrades | Berserk | Crit
Damage from passive levels

This is a very important DPS slot because it connects to **Weakened target damage** and **passive skill scaling**.

Base / rarity effects

Hatred:

On hit, inflicts Weaken on attacker, reducing its Attack by 10%.

ATK +15%

Defensive Stance:

Restores 15% of Energy Shield when skills level up.

ATK +25%

Potential Unleashed:

All passive skills learned in battle can be upgraded to Red Star, including character-exclusive passives.

Amplify:
+25% damage to Weakened targets.

Eternal forge path

E1:

Berserk:

+20 seconds Berserk when skill levels up.

+50% skill damage during Berserk.

Defeat monsters to extend Berserk duration.

Max 200 seconds.

E2:

ATK +10%

E3:

Wondermight:

The higher the shield value, the more damage dealt to Weakened targets.

For every 20% of Energy Shield value, +2% damage to Weakened targets.

E4:

Final HP +16000

E5:

Limit Break:

+40% Max Energy Shield.

Void forge path

V1:

ATK +10%

V2:

Stealth Attack:

+2% Crit Damage for every 2 levels of passive skills learned.

V3:

Final ATK +4000

V4:

Divine Might:

Initial passive skill level +1.

For each Red Star passive upgrade, Crit Damage +10%.

V5:

Final ATK +8000

Optimizer notes

This item should be scored very high when the build uses:

Weaken

High passive level count
Red Star passive upgrades
Crit Damage stacking
Skill Damage
Energy Shield

Database flags:

Slot = Gloves
DPSRelevant = Very High
Debuff = Weaken
ScalingInputs = PassiveSkillLevels, RedStarPassiveCount, EnergyShieldValue
StatTypes = SkillDamage, CritDamage, DamageToWeakened

4. SS Armor

Main association

SS Armor | Revival | Toughened Armor | Energy Orbs | Energy Level | Damage Multiplier | Skill Damage | Crit Damage

This armor has survival, but also real DPS through **energy level damage multipliers**.

Base / rarity effects

Revival Effect +35%
HP +15%
Toughened Armor:
Triggers Toughened Armor for 5 seconds when attacked.
60-second cooldown.
Immune to all damage during this period.
Collisions instakill minions.
HP +25%
Eternal Recharge:
Energy orbs drop every 50th kill.
Collect orbs to raise energy level, up to Lv.5.
+5% damage multiplier for each energy level.
Energy drains over time.
Super Enhance:
When total energy is above 90%, +30% additional skill damage.

Eternal forge path

E1:

Peerless:

-100 required hits for energy orb drops.

+5% additional damage multiplier for each energy level.

E2:

HP +10%

E3:

Suppression:

Able to trigger Super Enhance at Lv.4.

+5% additional damage multiplier for each energy level.

E4:

Final HP +16000

E5:

Ultimate Eternity:

-100 required hits for energy orb drops.

+10% additional Crit Damage for each energy level.

Void forge path

V1:

HP +10%

V2:

Cabalistic:

Energy level is maxed at the start.

Restore additional 2% of energy shield when looting energy orbs.

Shield damage boost +10%.

V3:

Final HP +16000

V4:

Destructive Restraint:

Energy drain speed halved.

Shield damage boost +20%.

V5:

Final HP +32000

Optimizer notes

This armor is not just tank gear. It creates a full **energy-level DPS system**.

Track:

EnergyLevel

EnergyOrbDropRequirement
DamageMultiplierPerEnergyLevel
SkillDamageWhenEnergyAbove90
CritDamagePerEnergyLevel
EnergyDrainRate
StartingEnergyLevel

Database flags:

Slot = Armor
DPSRelevant = High
ScalingInput = EnergyLevel
StatTypes = DamageMultiplier, SkillDamage, CritDamage
IgnoreForDPS = Revival/HP unless survival mode is added

5. SS Belt

Main association

SS Belt | Crit Rate ramp | Energy Shield restore | Eternal Laser | Crit Rate threshold | Crit Damage threshold | Skill Damage threshold

This is one of the most important optimizer items because it has **crit breakpoints**.

Base / rarity effects

Crit Rate +5% for every 500 effective attacks in battle.

Max 30%.

ATK +15%

0.5% of energy shield restored when a critical hit defeats a monster.

Max 5 times every second.

ATK +25%

Shield damage +10%

Light Judgment:

Fires 1 Eternal Laser every 10 seconds.

Marks 1 boss 10 seconds before firing and enters Charging Attack state.

During this time, 100% of total damage dealt to boss is converted into Charging Damage.

Charging Damage fires as a single laser when charge phase ends.

Charging damage is not affected by attack multiplier bonuses.

Eternal forge path

E1:

Mega Crit I:

If Crit Rate exceeds 100%, Crit Damage +30%.

E2:

ATK +10%

E3:

Mega Crit II:

If Crit Rate exceeds 130%, Skill Damage +30%.

E4:

Final ATK +4000

E5:

Mega Crit III:

If Crit Rate exceeds 150%, Crit Damage +100%.

Void forge path

V1:

ATK +10%

V2:

Revolution:

Shield damage boost +20%.

Restore 10% energy shield after laser is fired.

V3:

Final ATK +4000

V4:

Final Evolution:

During Eternal Laser charging period, Crit Rate +20%.

Stored Damage +30%.

V5:

Final ATK +8000

Optimizer notes

This belt should be a major DPS engine if the player can hit Crit Rate breakpoints.

Track:

EffectiveAttackCount

CritRateFromBelt

TotalCritRate

CritRateAbove100

CritRateAbove130

CritRateAbove150

EternalLaserChargeState

StoredDamage
BossMarked
ShieldRestoreOnCritKill

Database flags:

Slot = Belt
DPSRelevant = Very High
Breakpoints = 100% CR, 130% CR, 150% CR
StatTypes = CritRate, CritDamage, SkillDamage, StoredDamage
ModeValue = Very High for boss modes

This connects directly to the earlier **Crit Rate & Crit Damage** data type.

6. SS Boots

Main association

SS Boots | Movement Speed | Chilled enemies | Crystal Trail | Ice Blocks | Glacial Bloodline | Vulnerability | Damage stacks

The boots are very conditional. They are strong if the build can maintain chill/glacial stacks.

Base / rarity effects

Base Movement Speed +3
HP +15%
Crystal Trail:
Moving leaves a trail of ice blocks lasting 10 seconds.
Ignites key explosions when they expire.
Enemies touching blocks or hit by explosions are Chilled for 10 seconds.
Unstackable.
HP +25%
Damage to Chilled enemies +15%
Glacial Bloodline:
Move to start charging.
The faster the speed, the faster the charge.
Charge weakens when movement stops.
At full charge, gain 1 Glacial Bloodline stack.
Each stack gives +12% damage, lasting 15 seconds.
Max 5 stacks.

Repeat to extend duration.

Eternal forge path

E1:

Ice Armor:

2% Energy Shield.

+4% shield damage boost for each Glacial Bloodline stack.

E2:

HP +10%

E3:

Enhanced Ice Armor:

Shield damage boost +21%.

For every 25% shield HP gained, Movement Speed +10%.

E4:

Final HP +16000

E5:

Rime Blast:

Charge stacks +1.

Void forge path

V1:

HP +10%

V2:

Frost Blast:

Damage on Chilled enemies +20%.

V3:

Final HP +16000

V4:

Ice Arrow:

Damage on Chilled enemies +20%.

When crystal trail reaches 30 blocks in length, the trail immediately explodes, firing homing ice arrows.

Ice arrows inflict Chill and 10% Vulnerability for 10 seconds.

Unstackable.

V5:

Final HP +32000

Optimizer notes

Boots are good, but they are not always active. The calculator should score them based on whether the mode allows movement and whether chill uptime is realistic.

Track:

MovementSpeed
ChillUptime
DamageToChilled
GlacialBloodlineStacks
ChargeState
VulnerabilityFromIceArrow
ShieldDamageBoost
MovementRequired = True

Database flags:

Slot = Boots
DPSRelevant = High but conditional
Debuffs = Chill, Vulnerability
ConditionType = MovementBased / ChillBased / StackBased

Forge cost structure

The SS gear source also shows costs for each path.

The pattern is:

ForgePath
ForgeLevel
Relic/Core requirement
Faction core requirement
Fodder gear requirement

The image shows separate paths:

E1-E5 = Eternal path
V1-V5 = Void path
C1-C10 = Chaos path, shown for weapon

For the calculator, do not only store “effect.” Store **cost per upgrade** too, because relic cores are a major bottleneck.

Suggested fields:

ForgeCostID
GearSlot

ForgePath
ForgeLevel
RelicCoreCost
FactionCoreType
FactionCoreCost
FodderRarity
FodderCount
SourceImage

Important optimizer rule:

Relic cores should be spent where the next forge level gives the most DPS per relic core.

That means the recommender should compare:

DPSGain / RelicCoreCost
DPSGain / FactionCoreCost
DPSGain / FodderGearCost

Collectible sets tied to SS gear

At the bottom, the source shows SS gear collectible-set bonuses.

Erudite Heirloom

Associated mostly with **SS Weapon / Starforged Havoc**.

Each 3 yellow stars:

Starforged Havoc higher thrust damage with lower HP.

Max damage +20%.

Each 3 red stars:

Super Star E3:

Each Energy Overload stack's damage +10%.

25 stars total:

Disordered Death V2:

Targets take additional +15% damage.

Otherworld Treasure

Associated with **shield / Clarity / Reverse Scale**.

Each 3 yellow stars:
Omnicombat E5:
-5% required shield value to trigger each damage buff.
Each 3 red stars:
Clarity increases skill damage by +10%.
25 stars total:
Reverse Scale V2:
Triggered skill damage +20%.

Meaning of Life

Associated with **Fury / Weakened targets / Stealth Attack crit damage**.

Each 3 yellow stars:
When Fury is active, +10% additional skill damage.
Each 3 red stars:
+10% additional damage dealt to Weakened targets.
25 stars total:
Stealth Attack V2:
+1% additional Crit Damage for every 2 levels of passive skills learned.

Interdimension Movement

Associated with **SS Armor energy/shield system**.

Each 3 yellow stars:
Cabalistic shield damage boost +10%.
Each 3 red stars:
Suppression:
Super Enhance deals additional +5% Skill Damage every level.
25 stars total:
Eternal Recharge:
Each Energy level additionally adds +1% damage multiplier.

Optimizer notes

Collectibles should not be treated as separate random bonuses. They modify specific SS gear effects.

Database flags:

CollectibleSetBonus
AssociatedGearSlot

AssociatedForgeEffect
RequiredStars
BonusType
DPSRelevant

Most important DPS connections from this source

1. Crit Rate breakpoint system

The SS Belt creates major breakpoints:

Crit Rate >100% = Crit Damage +30%
Crit Rate >130% = Skill Damage +30%
Crit Rate >150% = Crit Damage +100%

This means Crit Rate is not linear. If an upgrade pushes the player over 100/130/150%, it becomes more valuable.

2. Debuff families involved

This SS gear source connects to multiple debuffs:

Weaken
Chill
Vulnerability
Void Fissure damage taken

That means this source must connect to your [Boss_Debuffs](#) table later.

3. Energy Shield / Energy Level are separate systems

Do not mix these up.

Energy Shield = shield HP system
Energy Level = armor/orb damage multiplier system

Energy Overload = weapon stack system

They sound similar, but they are different optimizer variables.

4. SS weapon, belt, gloves are probably the highest DPS modeling priority

For your calculator, the most urgent SS gear models are:

SS Weapon:

Energy Overload, Void Fissure, Chaos path, Void Power

SS Belt:

Crit Rate ramp, Crit thresholds, Eternal Laser

SS Gloves:

Weakened damage, Red Star passive upgrades, Crit Damage scaling

Armor and boots matter too, but they are more conditional.

Clean database format for this data type

SSGearEffectID

GearSlot

GearNameOrEffectFamily

ForgePath

ForgeLevel

EffectName

EffectDescription

AffectedStat

StatAmount

TargetScope

ConditionType

AssociatedDebuff

AssociatedMechanic

DPSRelevant

IgnoreForDPS

CostRelicCore

CostFactionCoreType

CostFactionCoreAmount

CostFodderType

CostFodderAmount

SourceImage

SourceDate

Example rows:

SS_WEAPON_E3_SUPER_STAR | Weapon | Starforged Havoc | Eternal | E3 | Super Star | Each Energy Overload stack damage increased to 35% | SkillDamage | Stack-based | CurrentBuild | EnergyOverloadActive | None | EnergyOverload | Yes | No | TBD | EternalCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

SS_BELT_E5_MEGA_CRIT_III | Belt | SS Belt | Eternal | E5 | Mega Crit III | If Crit Rate exceeds 150%, Crit Damage +100% | CritDamage | +100% | CurrentBuild | CritRateAbove150 | None | CritBreakpoint | VeryHigh | No | TBD | EternalCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

SS_GLOVES_V4_DIVINE_MIGHT | Gloves | SS Gloves | Void | V4 | Divine Might | Initial passive skill level +1; each Red Star passive upgrade gives Crit Damage +10% | CritDamage | +10% per Red Star passive | CurrentBuild | RedStarPassiveCount | None | PassiveSkillScaling | VeryHigh | No | TBD | VoidCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

SS_BOOTS_V4_ICE_ARROW | Boots | SS Boots | Void | V4 | Ice Arrow | Crystal trail explosion fires homing ice arrows that inflict Chill and 10% Vulnerability for 10s | Vulnerability | 10% | EnemiesHit | TrailLength30Blocks | Chill/Vulnerability | MovementTrail | High | No | TBD | VoidCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

Optimizer scoring rule

For SS gear recommendations, don't rank upgrades only by rarity. Rank by:

1. Does it increase direct DPS?
2. Does it unlock a crit breakpoint?
3. Does it add a debuff damage multiplier?
4. Does it improve boss damage specifically?
5. Does it scale with stats the player already has?
6. What is the relic core cost?
7. Is the condition realistic for the player's mode?

For your current damage-only style, the calculator should heavily favor:

Crit Rate breakpoint upgrades

Void Fissure / Vulnerability upgrades

Energy Overload stack upgrades

Damage to Weakened / Chilled targets
Skill Damage and Crit Damage scaling
Passive Red Star / passive level scaling
Boss laser / stored damage mechanics

1. Main chapter unlocks

From [Chapter_Unlocks-2024-09-17](#).

Chapter	Feature unlocked	What it does
1, after losing	Talent Tree	Permanent account upgrades using gold/oil-style progression. DPS value depends on ATK/crit nodes.
1, after losing	Shop	Basic item purchasing.
1, after losing	Equipment	Unlocks gear system: weapon, necklace, gloves, armor, belt, boots.
1, after winning	7 Days Challenge	Starter reward ladder.
1, after winning	Rookie Login Gifts	Starter login rewards.
1, after winning	Monthly + Super Monthly Cards	Paid/passive reward cards.
2, after losing	Growth Fund	Gem reward progression.
2, after losing	Daily Shop	Daily purchasable items.
2, after losing	Friends	Social/friend rewards.
2, after winning	Mission Control	Dailies and weeklies. Important for gem/pass/pass EXP income.
2, after winning	Main Challenges	Extra challenge content tied to chapters.

2, after winning	Quick Earnings	Idle rewards. Important for designs, gold, EXP.
2, after winning	Survivor Costumes	Cosmetic system; some may have stats depending on source.
2, after winning	Events	Limited-time events and shops. Big source of keys, shards, cores, collectibles.
3, after losing	Daily Rewards	Login/daily collection.
3, after winning	Clans	Unlocks clan system, clan shop, clan level rewards, clan expedition.
3, after winning	Daily Discount	Discount shop section.
4, after winning	Durian skill	In-battle skill unlock.
4, after winning	Laser Launcher skill	In-battle skill unlock.
5, after winning	Daily Event: Gold Mine	Gold farming mode.
6, after losing	Regular Challenge	Recurring challenge mode; also tied to Survivor Pass EXP.
6, after losing	Tech Parts after Regular Challenge	Unlocks tech part system. Important for Drone, Molotov, RPG, resonance.
7, after winning	Skill Visuals	Visual/cosmetic skill display feature.
8, after winning	Ender's Echo	Boss damage mode. Very important for DPS optimizer.
8, after winning	Survivor Pass / Super Survivor Pass	Paid/free pass reward systems.
8, after winning	Special Ops	Mission system for gems, gear, rewards.
9, after winning	Pets	Unlocks pet system, pet skills, pet awakening, pet toys.

9, after winning	Survivor level up	Unlocks leveling survivors for global stats.
10, after losing	Modular Mine	In-battle skill unlock.
10, after losing	Thunderbolt Bomb	In-battle skill unlock.
10, after losing	Inferno Bomb	In-battle skill unlock.
11, after losing	Permanent Privilege Card	Permanent card purchase. Important for long-term F2P/low-spend gem planning.
30, after winning	Zone Ops & Extreme Ops	Late challenge system. Useful for rewards and progression checks.

Big rule for the calculator: **chapter unlocks control what systems the optimizer is allowed to recommend**. Example: before Chapter 9, don't recommend pet awakening. Before Chapter 6, don't recommend tech parts.

2. Equipment system

Sources:

[Weapons_V2_2_2024-11-28](#)
[Necklaces_V2_2024-11-28](#)
[Eternal_Necklace_2024-11-28](#)
[Gloves_V2_2024-11-28](#)
[Armor_V2_2024-11-28](#)
[Belts_V2_2024-11-28](#)
[Boots_V2_2024-11-28](#)
[Merging_-_Equip_2024-11-28](#)
[Equip_Design_Costs_2024-11-14](#)
[Astral_Forge_2024-10-06](#)
[SS_Gear_2025-02-06](#)

What it does

Equipment is the main build system. It gives:

- Base ATK / HP
- Gear-specific effects
- Skill damage
- Crit rate
- Crit damage
- Boss damage
- Shield effects
- Revive effects
- Movement bonuses
- Debuff effects like Chill, Weaken, Vulnerability, Bleed, Laceration
- SS gear forge paths

Unlock requirement

- Equipment unlocks at **Chapter 1 after losing**.
- Merging requires duplicate gear and rarity progression.
- Astral Forge requires high-rarity gear and cores/materials.
- SS gear requires SS item creation/unlock plus relic cores/faction cores for upgrades.

Optimizer rule

For your calculator, equipment should be split into:

Normal Gear

S Gear

SS Gear

Astral Forge Upgrades

Collectible Gear Bonuses

Merge Cost / Upgrade Cost

HP-only gear effects should be stored but marked **DPSRelevant = No**.

3. SS gear system

Source: **SS_Gear_2025-02-06**

What it does

SS gear is the late-game gear layer. Each slot has special systems:

Slot	Main function
SS Weapon	Starforged Havoc, Void Power, Energy Overload, Void Fissure, Chaos Meteor, boss damage loops.
SS Necklace	Extra skill choices, skill damage, energy shield effects.
SS Gloves	Weaken, damage to Weakened targets, Red Star passive scaling, crit damage scaling.
SS Armor	Energy levels, energy orbs, damage multiplier, skill damage, crit damage by energy level.
SS Belt	Crit rate ramping, Mega Crit breakpoints, Eternal Laser stored damage.
SS Boots	Movement speed, Chill, Glacial Bloodline stacks, damage to Chilled enemies, Vulnerability from ice arrows.

Unlock / upgrade requirement

SS gear upgrades use:

Relic Cores

Eternal Cores

Void Cores

Chaos Cores

SS gear/fodder requirements

Forge path level: E1-E5, V1-V5, Chaos levels where available

Most important DPS breakpoints

From SS Belt:

Crit Rate >100% = Crit Damage +30%

Crit Rate >130% = Skill Damage +30%

Crit Rate >150% = Crit Damage +100%

This means the optimizer must value Crit Rate extra hard when it crosses one of those breakpoints.

4. Relic core usage

Sources:

`Relic_Core_Usage_-_Clan_Expedition_2024-11-29`
`Relic_Core_Usage_-_Clan_Expedition_mini_2024-11-29`
`Relic_Core_Usage_-_Enders_Echo_2024-11-29`
`Relic_Core_Usage_-_Enders_Echo_mini_2024-11-29`

What it does

These files show the recommended SS forge spread for different numbers of relic cores.

There are separate tables for:

Clan Expedition
Ender's Echo

because the best SS gear spread changes depending on mode.

Unlock / usage requirement

You need:

SS gear unlocked
Relic cores
Eternal/Void/Chaos cores
Enough faction cores/fodder

Optimizer rule

This becomes a relic-core recommender:

Input:

- Player relic core count
- Mode = Ender's Echo or Clan Expedition
- Current SS gear levels
- Crit rate
- Equipped gear

Output:

- Best next relic core spend

Important notes from the source:

- SS Gloves E3 requires a minimum of **101% Crit Rate from other sources, not including gloves**.
- For E5, you need about **121% Crit Rate** to have 100% uptime.
- The Clan Expedition table has special notes around **10, 29, 50 relic cores** and when to switch priorities.

So yes, the relic core data is already in your source pack.

5. Tech parts

Sources:

Tech_Parts_2024-10-09

Merging_-_Tech_Parts_2024-11-28

Tech_Part_Resonance_Cost_2024-11-28

Resonance_2024-10-30

Tech_Part_Resonance_Drone_2024-11-28

Tech_Part_Resonance_HI_Maintainer_RPG_2024-11-28

Twinborn_Part_2025-02-23

Unlock requirement

Tech parts unlock at:

Chapter 6 after losing Regular Challenge

What tech parts do

Tech parts modify in-battle skills. They turn normal skills into much stronger versions.

Major offensive tech parts include:

Tech part	Associated skill	What it does
Drone tech	Drones / Destroyer	One of the most important DPS techs. Unlocks Drone improvement and later Drone Resonance.

Lightning tech	Lightning Emitter / Supercell	Boosts lightning/electric skill damage and behavior.
Soccer tech	Soccer Ball / Quantum Ball	Improves soccer ball damage, duration, and behavior.
RPG tech	RPG / Sharkmaw Gun / C4 resonance	Improves rockets/explosions and becomes important in resonance.
Laser tech	Laser Launcher	Improves lasers and laser behavior.
Molotov tech	Molotov / Fuel Barrel	Improves fire pools, burn/fire effects, and Molotov resonance.

Major defensive/utility tech parts include:

Tech part	Associated skill	What it does
Brick tech	Brick / Dumbbell	Improves brick/dumbbell behavior and damage.
Guardian tech	Guardian	Improves knockback, guardian behavior, and damage.
Forcefield tech	Forcefield	Improves area field damage and utility.
Boomerang tech	Boomerang / Magnetic Dart	Improves boomerang behavior.
Durian tech	Durian	Improves durian movement/damage.
Rocket/defensive tech shown	Defensive skill modifiers	Some are more survival/utility than DPS.

Optimizer rule

For damage-only optimizer:

Drone tech = very high priority

RPG/C4 resonance = high priority if using that resonance path

Molotov resonance = high if fire/SF/poison/deep wound synergies are used

Defensive tech = only count if it gives real damage

HP/defense-only tech = store but ignore

6. Tech resonance

Sources:

[Tech_Part_Resonance_Cost_2024-11-28](#)

[Resonance_2024-10-30](#)

[Tech_Part_Resonance_Drone_2024-11-28](#)

[Tech_Part_Resonance_HI_Maintainer_RPG_2024-11-28](#)

What it does

Tech resonance lets a main tech part gain bonus effects from support tech parts. It adds:

Resonance Energy

ATK/HP from resonance

Special resonance skills

Multiplier upgrades

Drone wingbits

RPG/C4 explosion bonuses

Molotov/SF fire and poison bonuses

Resonance energy values

From [Tech_Part_Resonance_Cost](#):

Yellow tech part = 50

Yellow +1 = 100

Yellow +2 = 150

Yellow +3 = 200

Red = 300

Red +1 = 400

Red +2 = 550

Red +3 = 700

Red +4 = 850

Rainbow / resonance-tier part = 1000

Multiplier upgrade cost

1.0 → 1.2 = 1 resonance chip

1.2 → 1.4 = 1

1.4 → 1.6 = 2

1.6 → 1.8 = 3

1.8 → 2.0 = 5
2.0 → 2.2 = 5
2.2 → 2.4 = 7
2.4 → 2.6 = 9
2.6 → 2.8 = 12
2.8 → 3.0 = 15
Total = 60 resonance chips

Drone resonance

Drone resonance gives:

Attack Wingbit
Augment Wingbit
Attack Wingbit damage
Augment Wingbit damage
Wingbit interval reduction
Extra shots
Precision Device ATK bonuses
ATK/HP from resonance milestones

The special Drone source also says:

Augment Wingbit increases Destroyer/Drone damage.
Attack Wingbit attacks enemies with bullets.
Attack Wingbit base damage = x8 base attack.
Attack duration = 2s.
Base cooldown = 4s, can go down to 1s.
Augment Wingbit buff duration = 3s.
Base buff cooldown = 3s, can go down to 1s.

RPG / C4 resonance

RPG resonance is tied to:

RPG
Sharkmaw Gun
C4
Hi-Maintainer
Explosion radius
Launch interval
Explosion delay
C4 explosion damage

The C4 source says:

Base damage = x8 base attack

Maximum damage increase from resonance skills can be +240%

Base launch interval = 1.5s, max 0.6s

Throws new C4 every 1.5s

Base explosion delay = 4s, max 1s

C4 explodes after 4s from launch time

Molotov resonance

Molotov resonance is tied to:

Molotov

Fuel Barrel

SF / fire field effects

Deep Wound

Poison effect damage

Fire damage

SF generation interval

Optimizer rule

Tech resonance must be its own system. Do not just score it as “tech part rarity.”

Use:

MainTechPart

SupportTechParts

ResonanceEnergy

ResonanceMultiplier

ResonanceChipCost

SpecialMilestoneEffects

DPSGain

7. Survivor system

Sources:

Survivor_2024-09-07

Survivors_2025-01-26

Survivors_S_2025-01-26
Survivors_SP_2025-01-26
Survivor_Stat_Boosts_2024-12-12
Survivor_Upgrade_Costs_2024-12-19
Master_Yang_Skills_2024-11-28
Awakening_Core_Guide_2024-10-04

Unlock requirement

Survivor level-up unlocks at:

Chapter 9 after winning

What it does

Survivors give:

Active survivor skills
Passive skills
Global ATK / Crit / Crit Damage boosts
Star upgrades
Awakening skills
Teamwork passive slots
Mode-specific damage value

Survivor stat milestones

Important global milestone types:

Level 40
Level 80
Level 120
5 yellow stars
Red awakening stars

High-value DPS stats

ATK %
Crit Rate
Crit Damage
Skill Damage
Vulnerability

Laceration
Poisoned target damage
Chilled target damage
Weakened target damage
Teamwork passive slots

SP/TMNT survivors

TMNT survivors are SP-grade and event-limited.

Each 5-star TMNT survivor gives:

All Survivors Crit Rate +8%

That is huge for the optimizer because it feeds SS Belt crit breakpoints.

Awakening cores

Awakening cores are used for survivor awakening. The guide gives different priority paths for:

Ender's Echo
Clan Expedition
Yang
Metallia
King
Worm
Common
Catnips
TMNT survivors

Optimizer rule:

Awakening cores should be spent by DPS gain per core, not just favorite survivor.

8. Combat Harmony / Survivor Synergy

Source:

[Survivor_Combat_Harmony_Skills_and_Costs_2024-12-20](#)

What it does

Combat Harmony gives extra skill/passive effects and global bonuses.

It includes:

- Combat Harmony skill upgrades
- Survivor synergy levels
- Crit Rate bonuses
- Crit Damage bonuses
- Level cost requirements

Important DPS value

From the crit source and harmony source:

- Combat Harmony can give up to 30% global Crit Rate.
- Survivor Synergy can give up to 30% global Crit Damage.

Optimizer rule

This is a global account system. It should be treated like:

- GlobalCritRateSource
- GlobalCritDamageSource
- SurvivorInvestmentSystem

Very high long-term DPS value.

9. Crit Rate / Crit Damage system

Source:

[Crit_Rate_And_Damage_2025-03-09](#)

What it does

This tracks every major source of:

- Crit Rate
- Crit Damage
- Global crit

Conditional crit
Pet crit
Gear crit
Skill crit
Collectible crit
Survivor crit

Why it matters

Crit Rate does not only increase average damage. It also unlocks SS Belt thresholds:

>100% Crit Rate
>130% Crit Rate
>150% Crit Rate

Optimizer rule

Crit must be stored by source:

GlobalCritRate
GearCritRate
SkillCritRate
PetCritRate
CollectibleCritRate
ConditionalCritRate
GlobalCritDamage
GearCritDamage
SkillCritDamage
PetCritDamage
CollectibleCritDamage

Do not just make one flat crit number.

10. Debuff / multiplier system

Sources:

[Boss_Debuffs_2024-12-27](#)

[Shield_and_Skill_Damage_Multipliers_2024-12-27](#)

What it does

Tracks damage multipliers and enemy debuffs, including:

Chill
Poison
Exposed
Weaken
Debilitate
Yin Yang
Laceration
Vulnerability
Shield damage multipliers
Skill damage multipliers

Optimizer rule

Debuffs should be treated as their own damage layer.

Important reason: multiple systems say “damage to Chilled,” “damage to Weakened,” “damage to Lacerated,” or “Vulnerability.” Those only matter if the debuff exists and has uptime.

Use:

DebuffSource
DebuffDuration
DebuffStackLimit
DebuffUptime
DamageToDebuffedTarget
ModeReliability

11. Pet system

Sources:

[Pets_2024-11-28](#)
[Pet_Skills_2024-09-09](#)
[Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04](#)
[Pet_Awakening_Guide_2024-10-25](#)
[Pet_Awakening_Guide_-_Murica_2024-10-10](#)
[Pet_Awakening_Guide_-_Croaky_2024-10-11](#)

Pet_Awakening_Guide_-_Rex_2024-10-11
Merging_-_Pets_2024-11-28

Unlock requirement

Pets unlock at:

Chapter 9 after winning

What pets do

Pets give:

Direct pet damage
Owner Crit Rate
Owner Crit Damage
Owner Skill Damage
Pet Skill Damage
Pet Crit Rate
Pet Crit Damage
Pet cooldown reduction
Pet inherited ATK
Pet affection ATK
Assist skill bonuses
Toy bonuses

Important pets

Pet	Main use
Murica / Eagle	Direct pet DPS, whirlwind, penetration, cooldown, Crush.
Croaky / Frog	Owner skill damage, crit, long-term DPS support.
Rex / Dog	Owner crit rate/crit damage and long-term DPS support.
Crab	Strong pet DPS scaling: high attack, crit, cooldown, inherited ATK.
Robot	High theoretical damage with lasers/satellite, hit consistency matters.
Cat	Fast attacks, pet crit, cooldown, survival passive.
Fish	Utility/time stop, medium damage.

Turtle

Mostly defensive/tank.

Pet awakening requirement

Pet awakening uses awakening crystals and pet sacrifices.

Important cost path:

Y1 = 5 crystals

Y2 = 5

Y3 = 10

Y4 = 20

Y5 = 30

R1 = 40

R2 = 60

R3 = 60

R4 = 60

R5 = 60

Total = 350 crystals

Optimizer rule

The source says:

Take Rex or Croaky to R5 first.

After that, raise other pets in a balanced way to Y5 for better ATK per crystal.

For your damage-only calculator, pet HP and survival should be stored but ignored.

12. Collectibles

Sources:

[Collectible_Chest_Odds_2024-11-28](#)

[Collectible_Costs_2024-10-11](#)

[Collectibles_Edition_1_2024-11-28](#)

[Collectibles_Edition_2_2024-11-28](#)

[Collectibles_Edition_3_2024-11-28](#)

What they do

Collectibles give:

- Global ATK
- Crit Rate
- Crit Damage
- Gear-specific bonuses
- SS gear set bonuses
- Skill damage bonuses
- Target-specific damage bonuses

Unlock requirement

The source set has collectible chest odds and costs, but the exact chapter unlock is not shown in the chapter unlock chart. So in the optimizer, mark unlock as:

UnlockRequirement = Collectible system available / event or chest access

Do not invent a chapter until confirmed.

Optimizer rule

Collectibles must connect to exact gear/skill systems.

Example:

- Some collectibles boost SS Weapon Energy Overload.
- Some boost SS Belt/Armor/Gloves effects.
- Some give global Crit Rate or Crit Damage.

Collectibles are later-game but very important for final DPS.

13. Clan system

Sources:

[Clan_Shop_2025-02-17](#)

[Clan_Level_Rewards_2024-11-28](#)

[Clan_Expedition_Rewards_2024-11-28](#)

Unlock requirement

Clans unlock at:

Chapter 3 after winning

What clan does

Clan gives:

- Clan shop
- Clan expedition
- Clan level perks
- Clan member stat boosts
- Quick earning boosts
- Clan EXP rewards
- Clan currency spending

Clan shop

The clan shop source includes:

- S selector/chests
- S survivor shards
- Survivor shards
- Collectible keys
- Tech keys
- Pet/equipment related chests
- Relic/core-style items
- Other upgrade currencies

The chart shows stock changing by clan level:

- Level 1-11
- Level 12
- Level 13
- Level 14
- Level 15
- Level 16

Important: some clan shop items are icon-only in the image, so I'll store them by icon ID first if the exact item name is not readable.

Example:

ClanShopItem_01 | Cost 40000 | Qty 1 | Stock scales by clan level

Then later we map exact names.

Optimizer rule

Clan shop should be valued by:

Gem-equivalent value

DPS relevance

Rarity bottleneck

Player progression stage

Clan currency cost

Monthly/weekly stock limit

High-priority shop items are usually rare bottleneck items like relic cores, S selectors, shards, tech keys, collectible keys, and resonance/awakening materials depending on stage.

14. Survivor Pass

Sources:

[Survivor_Pass_2024-12-19](#)

[Survivor_Pass_Experience_2024-09-16](#)

Unlock requirement

Survivor Pass unlocks at:

Chapter 8 after winning

What it does

Survivor Pass gives reward tracks for:

Survivor resources

Tech parts

Equipment resources

Joey shards

Essence

Survivor shards
Reset vial key
Awakening core
Tech keys
S keys
Relic core
Collectible key

Pass EXP sources

From the pass EXP chart:

Dailies
Defeat monsters
Clear main chapters/challenges
Regular Challenge
Path of Trials
Ender's Echo
Clan Expedition
Special Ops

Season max shown:

15,590 to 15,740 EXP range

Optimizer rule

Pass value should be treated as:

RewardValue / Cost
DPSMaterialValue
RelicCoreValue
ShardValue
KeyValue

15. Gems, oil, essence, designs, item values

Sources:

Sources_Of_Gems_2024-10-07
Sources_of_Gems_2024-12-18
Sources_Of_Oil_2024-11-03
Sources_of_Oil_2024-12-18
Energy_Essence_Cost_2024-11-28
Energy_Essence_Cost_Full_2024-11-28
Equip_Design_Costs_2024-11-14
Item_Values_2024-10-02

What they do

These are economy/value sources.

They track:

Where gems come from
Where oil comes from
Survivor level essence costs
Equipment design costs
Item gem-equivalent values
Upgrade costs

Optimizer rule

This is needed for event/shop decisions.

Use it to answer:

Is this worth buying?
Should I spend gems?
Should I save for Black Friday?
What item has better gem value?
What is my bottleneck?

For your account, this matters because you are trying to avoid wasting gems and maximize DPS.

16. Ender's Echo rewards

Source:

EE_Rewards_2024-09-15

Unlock requirement

Ender's Echo unlocks at:

Chapter 8 after winning

What it does

Ender's Echo is a boss damage/ranking mode. It gives rewards based on performance/ranking.

Optimizer rule

For Ender's Echo, favor:

Boss DPS
Crit Rate breakpoints
Relic core spread from EE guide
Debuff uptime
Single-target damage
Pet direct DPS
Yang/Metallia/Joey mode-specific paths

Do not overvalue mob-clearing effects for EE unless they directly boost boss damage.

17. Skills

Source:

Skills_2024-11-28

What it does

This contains in-battle skills and likely their evo/passive pairings.

Skills are needed because tech parts, drones, resonance, survivor skills, and SS gear modify them.

Unlock requirement

Some skills are chapter-unlocked:

Durian = Chapter 4 after winning

Laser Launcher = Chapter 4 after winning

Modular Mine = Chapter 10 after losing

Thunderbolt Bomb = Chapter 10 after losing

Inferno Bomb = Chapter 10 after losing

Optimizer rule

Skills should be stored as:

SkillName

EvoName

PassiveRequired

BaseFunction

TechPartAssociation

ResonanceAssociation

DPSRole

MapClearValue

BossValue

18. Wishlist probabilities

Source:

[Wishlist_Probabilities_2024-11-28](#)

What it does

Shows odds/probabilities for wishlist pulls.

Unlock requirement

Requires the relevant chest/wishlist system to be available.

Optimizer rule

Use this for:

Expected value

Chest opening strategy

How many keys/gems needed

Chance to get target gear

19. Timed Muster medal conversion

Source:

[Timed_Muster_Medal_Conversion_2024-12-12](#)

What it does

Shows conversion/value for timed muster medals.

Optimizer rule

Use for event value and whether to convert/spend medals now or save/use differently.

20. Master source index for the calculator

This is the source pack I'll use going forward:

Chapter_Unlocks

Skills

Boss_Debuffs

Shield_and_Skill_Damage_Multipliers

Crit_Rate_And_Damage

Survivor

Survivors

Survivors_S

Survivors_SP

Survivor_Stat_Boosts
Survivor_Upgrade_Costs
Survivor_Combat_Harmony_Skills_and_Costs
Master_Yang_Skills
Awakening_Core_Guide

Weapons
Necklaces
Eternal_Necklace
Gloves
Armor
Belts
Boots
SS_Gear
Astral_Forge
Relic_Core_Usage_Enders_Echo
Relic_Core_Usage_Clan_Expedition

Tech_Parts
Tech_Part_Resonance_Cost
Resonance
Tech_Part_Resonance_Drone
Tech_Part_Resonance_HI_Maintainer_RPG
Twinborn_Part
Merging_Tech_Parts

Pets
Pet_Skills
Pet_Awakening_Costs_and_Toy_Bonuses
Pet_Awakening_Guide
Pet_Awakening_Guide_Murica
Pet_Awakening_Guide_Croaky
Pet_Awakening_Guide_Rex
Merging_Pets

Collectible_Chest_Odds
Collectible_Costs
Collectibles_Edition_1
Collectibles_Edition_2
Collectibles_Edition_3

Clan_Shop
Clan_Level_Rewards
Clan_Expedition_Rewards

Survivor_Pass
Survivor_Pass_Experience
EE_Rewards

Sources_Of_Gems
Sources_Of_Oil
Energy_Essence_Cost
Energy_Essence_Cost_Full
Equip_Design_Costs
Item_Values
Merging_Equip
Wishlist_Probabilities
Timed_Muster_Medal_Conversion